School Sports Management System for Kg/ St. Joseph's Balika Vidyalaya.

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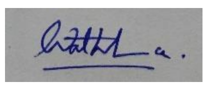
Academic Year 2020

This dissertation is submitted in partial fulfillment of the requirement of the Degree of Bachelor of Information Technology (external) of the University of Colombo School of Computing

DECLARATION

"I certify that this dissertation does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma in any university and to the best of my knowledge and belief, it does not contain any material previously published or written by another person or myself except where due reference is made in the text. I also here by give consent for my dissertation, if accepted, to be made available for photocopying and for interlibrary loans, and for the title and abstract to be made available to outside organizations."

Signature of candidate:  Date: 2021/01/28

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ACKNOWLEDGEMENT

I take this opportunity to express my gratitude to everyone who helped me and guided me to complete to the project successfully.

I express my sincere thanks to the respected personnel of the University Of Colombo School Of Computing, for giving me the opportunity to do this project work.

I wish to thank my project supervisor Mr. M.P. Sarath Wijesinghe, who gave me the necessary technical guidance and constant supervision despite his busy schedule.

I’m very grateful to Mrs. Mallika Ranasinghe Principal of Kg/St. Joseph’s Balika Vidyalaya, for permitting me to do this project and I am also grateful to Deputy Principals and teachers of the school, who enlightened me about the process and functionalities by providing necessary information and documents regarding the sports management

I would also like to thank Esoft Metro Campus and respectable teaching panel for providing valuable knowledge and assistance throughout the BIT degree program.

I am also grateful to my parents, husband and all my family members for all the support and encouragement they gave me to complete this project successfully.

ABSTRACT

St. Joseph's Balika Vidyalaya is one of leading ladies' school. This leadership is lighted via not only academic achievements but also various extra-curricular activities. As the main extra-curricular activity, sport has achieved to the national and the international level. The school maintains a manual system for the sports management. And the system is not fully efficiency and effectiveness. The processes of transaction and communications with external parties, keeping records, retrieving records when required, are very time consuming. The time is gold in the teaching learning process, thus there is a huge challenge to the student as well as the teacher to balance the education and the sport. The school cannot face this challenge successfully by having the current manual system. Converting the manual system into a computerized system is the best solution for avoiding these circumstances. A Computerize system is more efficient and effective than current manual system.

New System was developed by breaking down the requirements into few main modules namely User management, Game and event management, User registration, Tournament management, Practice session management, Notification and Report generating. Each and every module is targeted the specific group of functionalities needed to fulfill the final outcome of the system. The administration module handles the system related functions whereas other modules handle the sports management related functions. Student registration module let the staff to handle and maintain student registrations. Report generating module let the users to generate few types of sports related reports. Notification module let system users to know notices.

The system has been developed using PHP which contains the core programming language; JavaScript has been used to client-side validations. MySQL has been used to manage the databases and Apache web server has being used to implement the system. XAMPP development kit has been used throughout the project development.

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CHAPTER 01- INTRODUCTION

1.1. BACKGROUND OF THE PROJECT

According to Theodore Hesburgh, author of "The Importance of School Sports and Education," it is imperative for school age children to have access to sports and games. Not only does it empower youth and promote higher self-esteem, it also motivates students, enables them to earn better grades, especially in schools where obtaining certain grades is a pre-requisite to staying on the team. Numerous physical benefits include maintaining a healthy weight, preventing chronic diseases and learning the skills necessary to maintain a healthy lifestyle after graduating. [1]

It seems that the importance of sports in student life. This project is developed for the Kg/ St. Joseph's Balika vidyalaya to handle their sports related activities effectively and efficient way. It has been formed number of national and international level sportswomen since long years.

In the present system all the functions of sports are maintained manually. Due to increased number of records writing into books and searching whenever it is required is a time consuming as well as complicated job.

Nowadays information and communication technology is one of the rapidly increasing filed in the world. ICT is a very important tool in the education sector and there are various uses of ICT. Some of these include administration, L/T process, co-curriculum activities and communication. Different parties use various ICT related procedures to perform their day to day task for instance generate reports, send information to communities.

Accordingly building up a computerized system to replace the manual system is an important to archive school effectiveness into a maximum level.

1.2. MOTIVATION FOR THE PROJECT

Education is incomplete without sports and now-a-days sports are the integral part of the education. In schools it, the children are taught some sorts of games in very early stage to keep their value in life. Sports are also a part of academic curricula. The research proves that in a public-school classroom half of the students are overweight. There is a lot of improvement in poor food quality, culture of over-eating, and inactive lifestyles. Hence sports education is very much essential for today's youth generation.

As the leading school in Kegalle district Kg/ St. Joseph's B.V. is conducting large numbers of various types of sports to develop students' physical and mental robustness. The management of the school keeps up all the sports related works manually. Due to increased number of records writing into books and searching whenever it is required is a time consuming as well as complicated job. Data duplication is available and space is wasted. Such manual systems' data may be damaged or destroyed due to unavoidable circumstances. Informing notices to parent via students some may cause doubtful. Selecting students for school color awards without any mistake is become a major challenge to the school management.

Nowadays most of whichever fields tend to use ICT supported systems in order to do operations methodically. A convenient sport management system will improve the quality of the sport and it will enhance the relationship between all the parties bounded surrounding sports of the school.

This computer-based sports management system will facilitate to reduce the issues and to provide a better efficient sports management to the school.

1.3. OBJECTIVES AND SCOPE OF THE SYSTEM

- Login Registration

  Admin provides user names and passwords to other system users. He also has the right to modify or delete the given user names and passwords.

- Create games and events.

  System should have details of games and its events to perform the system activities.
- Assign PTI and teacher for each game.
- Student Registration

  The system allows registering students who wishes to join a sport team or play some game. Here the P.T.I. is the responsible actor to enters all the details of students. All this information will be stored in a database.
- Students' BMI value calculation.

  BMI value of each student will be calculated periodically to monitor their physical fitness. And own BMI value can be viewed via students' profiles.
- Upcoming Tournaments and Game Management

  Upcoming tournament details are saved in a database and it will be visible to all system users via a calendar option.
- Select students for teams or individual events.

  Coach and the PTI has the permission to select students by detecting recorded performances of selection meets and checking physical fitness of them.
- Request medical certificate.

  Some events require a medical certificate to participate tournaments. Those who participate that kind of events are informed electronically to complete the medical test. If not list of MC.s required students can be generated to transfer them to the medical officer.

- Maintaining the student sports achievements electronically.

  System allows insertions, retrievals, deletions and updating of the students' sports achievements and photos.
- System has eligibility to nominate students for school color awarding, based on their achievements.

- Practice sessions attendance marking

  Attendance of the students for the practices can be marked through the system.
- Students sports information dissemination and notification management

  System allows students to view their sports' performances and the notifications.

- File uploading

  Sports' coaches can upload their filled payment voucher to the system to get their monthly payments on behalf of hand it over manually.

- Handles notifications

  The parents will receive relevant notices electronically on time. It will save time of the both parties by avoiding unnecessary long telephone calls, printing paper cost and posting cost.
- Generates reports

  Monthly attendance, sports achievements and eligible lists of color awardees can get as reports.
- Back up information.

  System information should be backed ups when it’s essential to do.

1.4. STRUCTURE OF THE DISSERTATION

There are six main chapters in the dissertation to bring about an overall comprehension on the School Sports Management System.

Introduction chapter defines what the project is, motivation for the project, scope and objectives of the project.

Analysis chapter explains about existing system, user requirements and feasibility study to make it easier to understand the system.

Design chapter includes some diagrams relating to the system design, databases and user interfaces.

Implementation chapter explains how the system is implemented in user environment, Hardware software requirement, and main code segments of the system. The test plan and test cases of the system are given in the

Evaluation chapter. The final chapter, the

Conclusion includes lessons learnt and the information about further development of the system.

CHAPTER 02- ANALYSIS

2.1. INTRODUCTION

"System Analysis is the process of studying a procedure or business in order to identify its goals and purposes and create systems and procedures that will achieve them in an efficient way." [2]

It seems the System Analysis is the first phrase and a very important phase of the software development life cycle. The systems analysts are responsible for identifying requirements of the system.

2.2. EXSISTING SYSTEM

In the present system all records are maintained manually. When a student registers to a sport by filling the registration application his/her name and other details are collected into a record file and issue a register number. This duty is done by teacher in-charge of each game. Due to increased number of records writing into book and searching whenever it is required is a tidy job. Data duplication is available and space is wasted.

All the student records are kept in manually. When a student or team wins an event or a tournament, the teacher in charge has to note down all the records in a record book. It can be damage due to any natural disaster or any displacement.

The process of nominating students into colour awards has many steps such as form filling, signing for recommendations, attaching copies of certificates, checking record books to confirm. Teacher in charge has to turn the pages and fill the relevant applications manually. It is a very time consuming and a bored process.

Students' attendance information is not maintained properly for each game and difficult to generate monthly reports. Informing notices via paper documents is not a cost-effective process. And also, it is an untrustworthy process. There is no proper way of taking backups of the important information such as details of participated tournaments.

When we consider the manual system, it has many draw backs rather than benefits.

2.3. USECASE DIAGRAM FOR THE EXSISTING SYSTEM

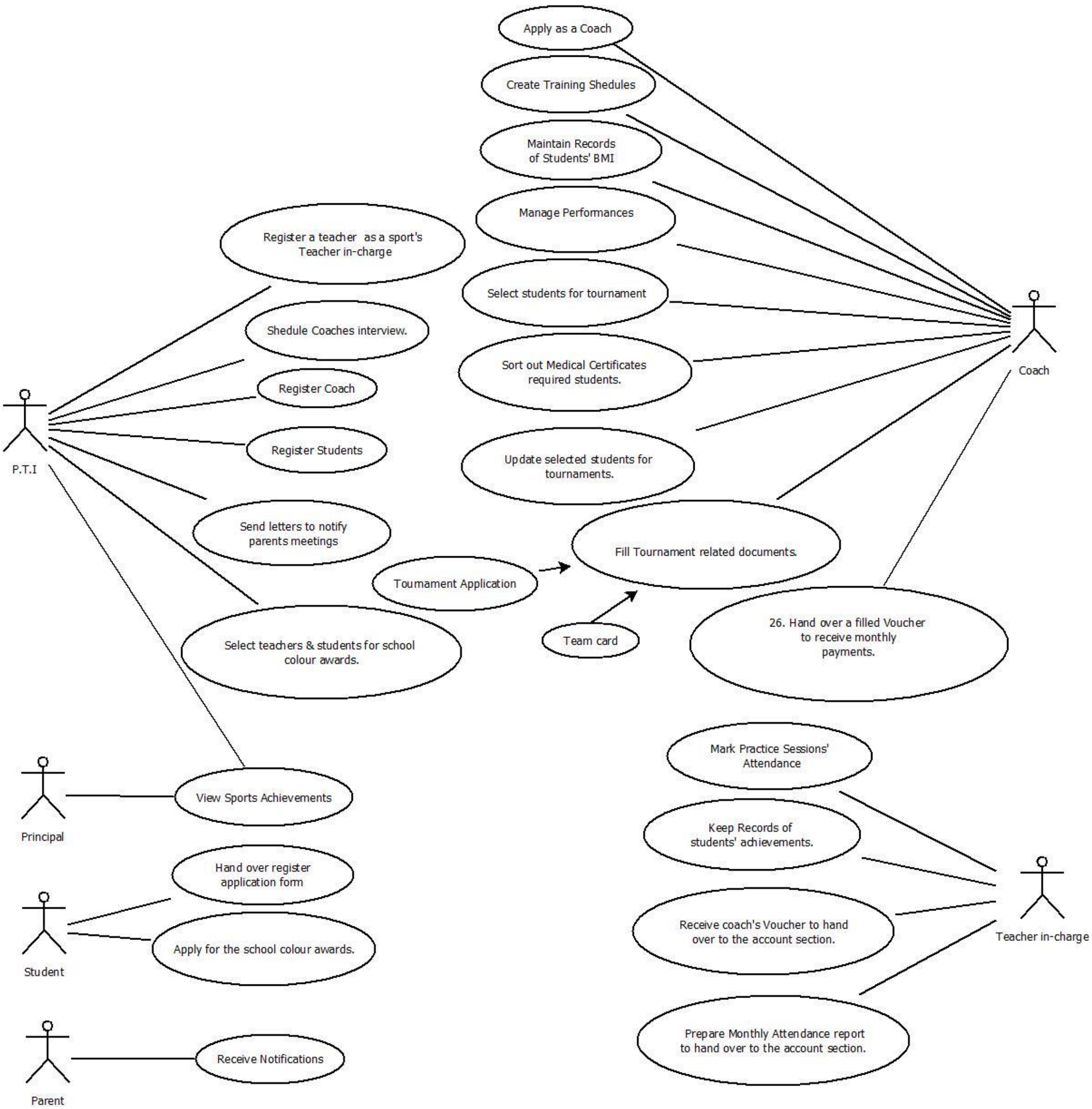


Figure 2.1: Use Case Diagram for the Existing system

2.4. FUNCTIONAL REQUIREMENTS

- User management module should allow,
    - Administrator to create user and user roles.
    - Administrator to assign user permissions.
    - Administrator to activate/deactivate users.
    - User to login to the system using their username and password and verify username and password.
    - User to log out from the system.
    - Destroy the session when logged out the user.
    - System to update user profiles.

- Games and Events management module
    - Principal to create games.
    - Principal to update games.
    - Principal to activate/ deactivate games.
    - PTI to add events.
    - PTI to update events.
    - PTI to activate/ deactivate events.

- User registration module should allow,
    - Principal to assign PTI and teachers for the games.
    - Principal to schedule interview for recruiting game coaches.
    - Principal to interview applicants for the post of Game's Coach.
    - Principal to register coach.
    - Principal to update PTI, Teacher in-charge, coach.
    - Principal to activate/ deactivate PTI, Teacher in-charge, Coach.
    - P.T.I. to register student.

- Tournament management module should allow,
    - Principal to add upcoming tournaments.
    - Coach to select students for tournaments.
    - Coach to sort medical certificate required
    - Teacher in-charge to record participated tournament information. (Achievements)
    - Teacher in-charge to update participated tournament information. (Achievements)
    - All users to view students' achievements under selected criteria.
    - PTI to nominate students for school color awards.

- Practice sessions management module should allow,
    - Coach to schedule practice sessions.
    - Coach to inform scheduled practice sessions to PTI, Teachers in-charge and students.
    - Coach to update scheduled practice sessions.
    - Coach to cancel scheduled practice sessions.
    - Teacher in-charge to mark attendance of students.
    - Coach to calculate students' BMI

- Notification module should allow
    - System to send remainder to remind upcoming tournament dates.
    - PTI to inform special notices for parents.

- Report Generation module should allow,
    - PTI to generate filled tournament application, team card, list of medical certificates required students and list of color awardees.
    - Teacher in-charge to generate monthly attendance sheets for training sessions.
    - Principal and PTI to generate reports of students' achievements according to select criteria.
    - PTI and teacher in-charge to print the reports and documents

2.5. NON-FUNCTIONAL REQUIREMENTS

- User friendliness

  Student profiles and the system are designed as simple and interactive to allow handling easily by the users. It means the learning curve of the new systems must be relatively low.
- Reliability

  This is the trustworthiness of all the stakeholders and between the systems.
- Usability

  Usability means ease of use the system components.
- Security

  Sensitive personal information are only visible to own via own profiles.
- Maintainability

  Customer profiles and various value changes should be update when they occurred.

2.6. SIMILAR SPORTS MANAGEMENT SYSTEMS

Before it was decided to develop the proposed School Sports Management System several existing alternative systems and web sites were considered. The alternative systems and websites are as follows.

MES Athletic Director

Figure 2.2 MES Athletic Director

MES Athletic Director is a full-featured software program designed to do the paperwork part of your job and allow you to manage your sports program with a minimum of effort. It contains feathers of Rosters, Schedules, Contracts and eligibility, Track contracts sent/received, Home/Away game reports, Class release reports, Game reminders for opponent sand officials Monthly and weekly event calendars, Mailing labels, Transportation requests, Budget features Award certificates. [3]

CHAPTER 03: DESIGN

3.1. INTRODUCTION

This chapter describes alternative design solutions and its justifications for the propose project, high level systems architecture, design techniques and design decisions, models and methodologies that are used. It also shows how user interfaces and databases have been designed, approach that has been used.

3.2. PROCESS MODEL

"A software process (also knows as software methodology) is a set of related activities that leads to the production of the software. These activities may involve the development of the software from the scratch, or, modifying an existing system." [4]

Such models are Waterfall Model, Prototyping Model, Rapid Application Development (RAD), and Rational Unified Process (RUP)

3.3. PROCESS MODEL FOR PROPOSED SYSTEM

Among the process models, waterfall model was selected for the proposed system by considering its sequential software development process. Some advantages of this model are listed below.

- Uses clear structure when compared with other methodologies.
- Waterfall focuses most on a clear, defined set of steps.
- Determines the end goal early.
- Transfers information well.

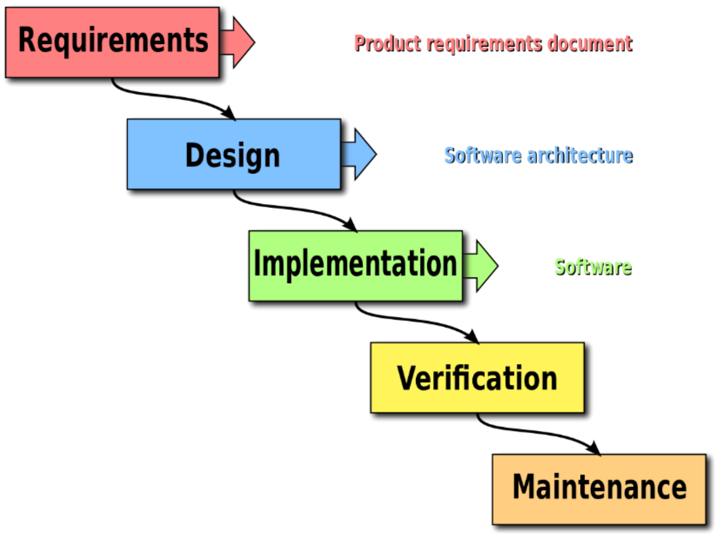


Figure 3.1 – Phases of Waterfall model

The main phases of Waterfall model are described below.

Analysis

The product development team analyzes the requirements, and fully understands the problems. This is a research phase that includes no building. The team attempts to ask all the questions and secure all the answers they need to build the product requirement.

Design

The software developers design a technical solution to the problems set out by the product requirements, including scenarios, layouts and data models. This phase is usually accompanied by documentation for each requirement, which enables other members of the team to review it for validation.

Implementation

Once the design is approved, technical implementation begins. This is often the shortest phase because research and design have been done in advance.

Testing

Upon completion of full implementation, testing needs to occur before the product can be released to customers. The software testing team will use the design documents, personas and user case scenarios delivered by the product manager in order to create their test cases.

3.4. HIGH LEVEL SYSTEM ARCHITECTURE

The system will be developed mainly focusing on procedural and object-oriented development techniques. PHP will be used as the programming language. MySQL will be used to handle database querying and related communications. This web-based system will be running on Apache HTTP Web Server.

3.5. THE DESIGN TECHNIQUES

Object oriented techniques are the most preferred and widely used standard of developing and designing a system. Because it promotes code reuse, code maintainability, modularity, etc. Accordingly, object-oriented design techniques such as use case diagrams using UML notation will be used in the design stage. But still there are some tools/ libraries does not totally compatible with object-oriented approaches. Accordingly, it was difficult to totally move with object-oriented techniques throughout the development cycle. Accordingly, it was decided to use both object-oriented techniques and structured approaches in development stage vice versa where applicable.

3.6. USE CASE DIAGRAM FOR THE COMPUTERIZED SYSTEM

“A use case is a software and system engineering term that describes how a user uses a system to accomplish a particular goal. A use case acts as a software modeling technique that defines the features to be implemented and the resolution of any errors that may be encountered.” [5]

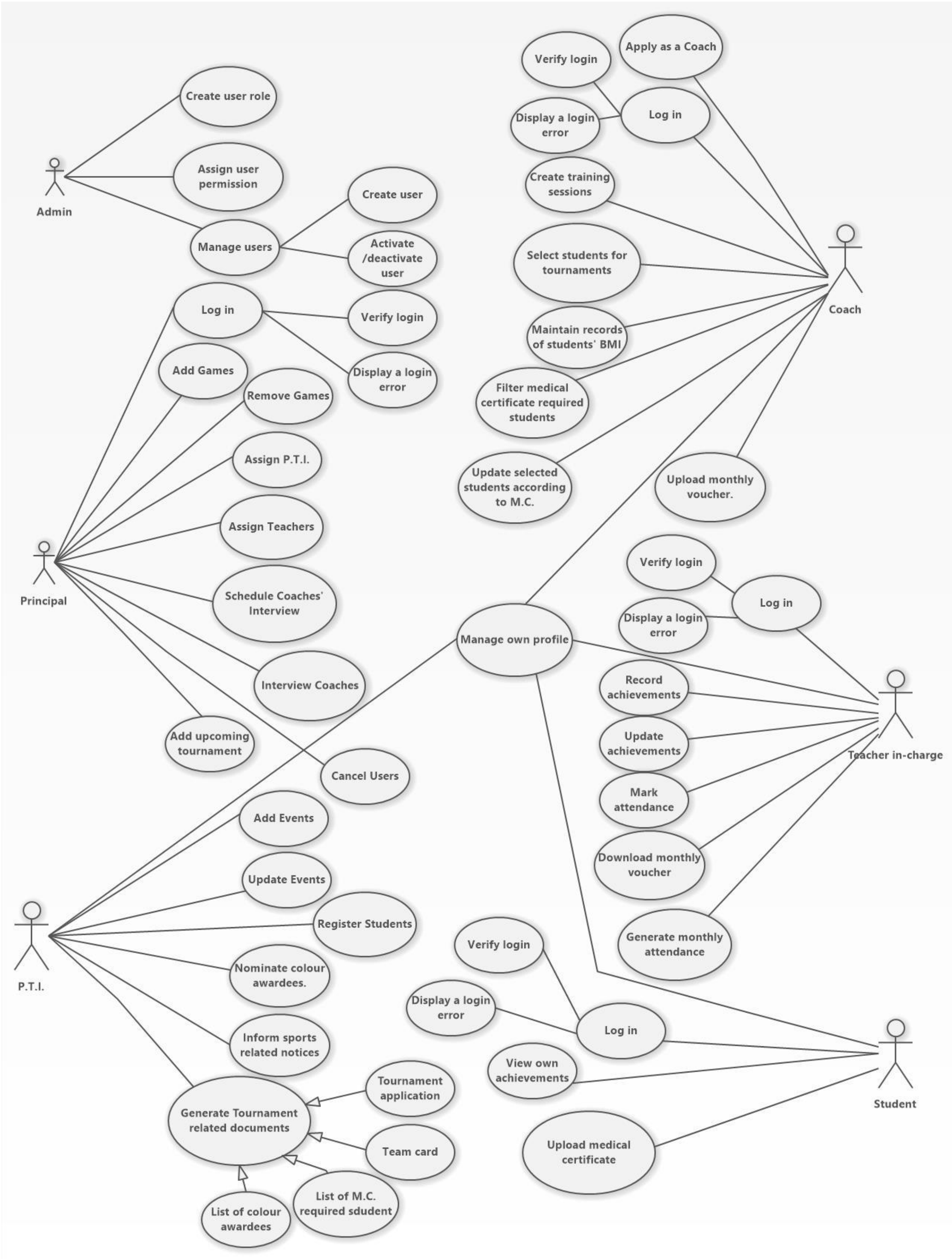


Figure 3.2: High Level Use Case Diagram for the Computerized System

3.7. NARRATIVE USE-CASES

Use Case Login

<table>
<tr><td>Use case number</td><td colspan="2">UC-01</td></tr>
<tr><td>Use-case Name</td><td colspan="2">Login</td></tr>
<tr><td>Priority</td><td colspan="2">High</td></tr>
<tr><td>Actor</td><td colspan="2">Admin, PTI, Coach, Teacher in-charge, Principal, Student</td></tr>
<tr><td>Description</td><td colspan="2">This use case describes how system users are log into the Sports Management System.</td></tr>
<tr><td>Pre-condition</td><td colspan="2">None</td></tr>
<tr><td>Post condition</td><td colspan="2">If the use case was successful, the actor is now logged into the system. If not, the system state is unchanged.</td></tr>
<tr><td rowspan="2">Basic course of Action</td><td>User Action</td><td>System Response</td></tr>
<tr><td>1. The Actor is on the home page to login to the system.

3. The Actor enters username and password, click on the Login Button.</td><td>2. The system promotes the Actor to enter Username, Password.
4. The system verifies that all the filled have been filled out and valid.
5. The system successfully logged in the system.
6. Use case Exit</td></tr>
<tr><td>Alternate course of Action</td><td colspan="2">4.1 If all fields are not filled out and not matched to the username and password the system notifies the actor a message “Invalid Username or Password” and then goes back or returns to step 3 of basic course of Action to enter again.</td></tr>
</table>

Table 3.1Use case: Login

Use case Schedule Coaches' Interview

Use case number	UC-07	
Use-case Name	Schedule Coaches' interviews.	
Priority	High	
Actor	Principal	
Description	This use case permits Principal to schedule Coaches' interviews.	
Pre-condition	UC- 01, UC-05	
Post condition	Principal successfully schedules Coaches' interviews.	
Basic course of Action	User Action	System Response
	1. Principal wants to schedule Coaches' interviews. 2.Actor requests the page of 'Schedule Interviews'. 3. Actor clicks on 'View Applicants' button. 5. Principal clicks on the 'Schedule interview' button. 7. Principal enters the following information to the form. (Start Date, End Date, Starting time, Time Duration) 8.Principal clicks on 'Ok' button. 12. Principal clicks on 'Send schedule' button.	4.System displays the number of applicants for each game. 6. System displays a form to enter schedule information. 9. System verifies that all the fields are filled. 10. System verifies that the filled data is valid. 11. System completes schedule and displays a message as 'Interview schedule is successfully completed.' 13. System send the schedule to all the applicants via email. Use case Exit
Alternate course of Action	If one or few fields are empty the system back to basic course of action 4 and display a message as '<Field> should not be empty.' 10.1 If the entered information is invalid the system back to basic course of action 4 and display a message as '<Field> is invalid'.	

Table 3.2: Use case: Schedule Coaches' Interview

Use case Athletic Skill Test.

Use case number	UC-07	
Use-case Name	Athletic Skill Test	
Priority	High	
Actor	Coach	
Description	This use case permits Coach to divide athletes into events.	
Pre-condition	UC- 01	
Post condition	Coach successfully divides athletes into events.	
Basic course of Action	User Action	System Response
	1. Actor wants to divide athletes into events. 2.Actor requests the page of 'Athletic Skill Test'. 3. Actor clicks on relevant link 5. Actor enter records.	4.System displays relevant events under selected event type and name of athletes. 6. System automatically saves records. 7. System divide students into events 8 System displays a message as 'Action Success.' 9. Exit use case.
Alternate course of Action	5.1 If the entered information is invalid the system back to basic course of action 4 and display a message as '<Field> is invalid'.	

Table 3.3 : Use case: Athletic Skill Test

Use case Register student

Use case number	UC-09	
Use-case Name	Register student	
Priority	High	
Actor	Teacher in-charge	
Description	This use case permits Teacher in-charge to register students.	
Pre-condition	UC-01	
Post condition	Register student successfully.	
Basic course of Action	User Action	System Response
	1. The teacher wants to register a student. 2.Teacher requests the page of Register Student. 4. Teacher enters the following information to the form. (Date, Register year, Name with initials, DOB, Full Name, Admission no, Address, Parent Mobile Number, Parent Email address, NIC Number, Photograph, Game) 5. Teacher clicks on the 'Register Student' Button.	3.The system displays the form to be filled to register student. 6. The system verifies that all the fields are filled. 7. The system verifies that the filled data is valid. 8. System displays a message as 'Student is registered Successfully.' 9. Use case Exit
Alternate course of Action	6.1 If one or few fields are empty the system back to basic course of action 4 and display a message as '<Field> should not be empty.' 7.1 If the entered information is invalid the system back to basic course of action 4 and display a message as '<Field> is invalid'.	

Table 3.4: Use case: Register Student

3.8. ACTIVITY DIAGRAMS

"In UML an activity diagram is a graphical representation of an executed set of professional activities and considered a state chart diagram variation. Activity diagram describes parallel and conditional activities. Use cases and system functions at a detailed level." [6]

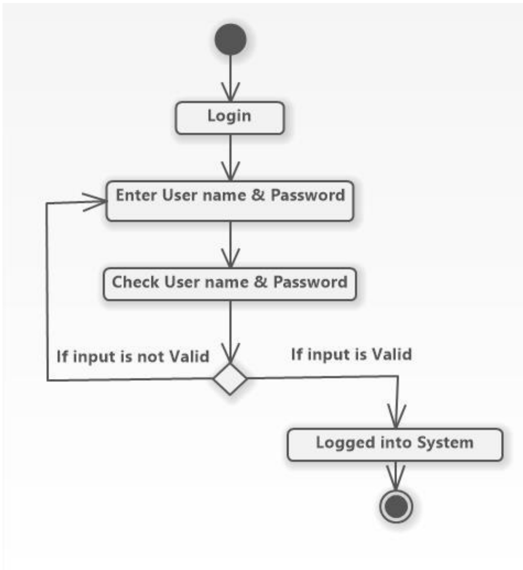


Figure 3.4: Activity Diagram for system login.

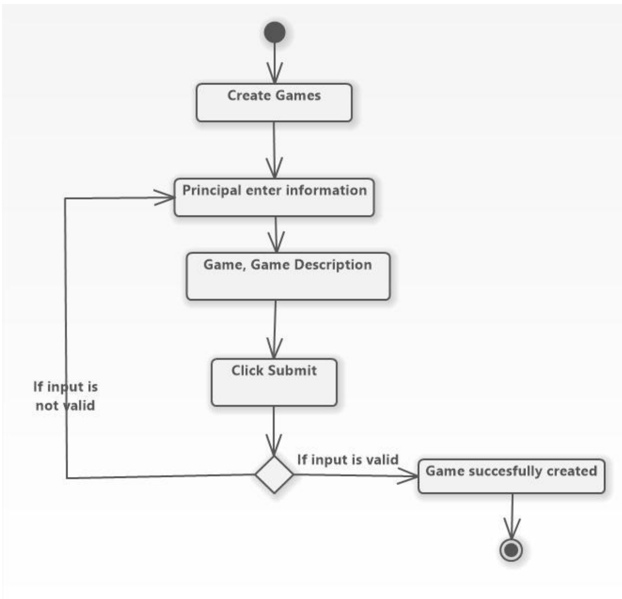


Figure 3.5: Activity Diagram for the creating games.

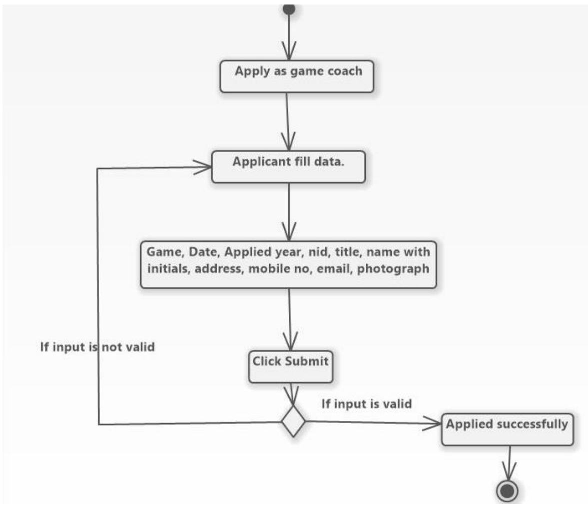


Figure 3.6: Activity Diagram for the Apply as coach

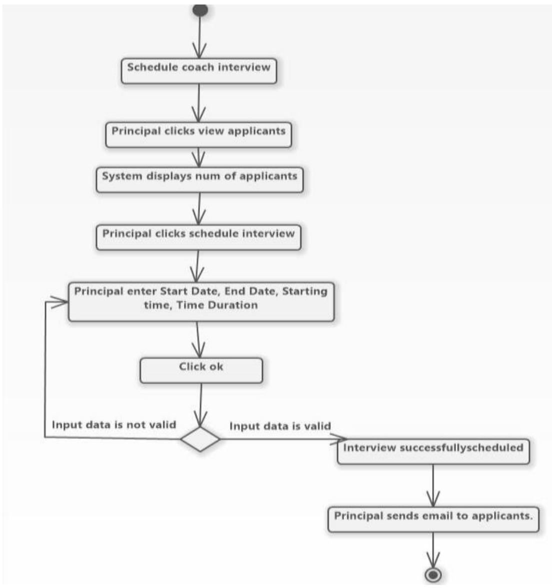


Figure 3.7: Activity Diagram for the Coach **interview**

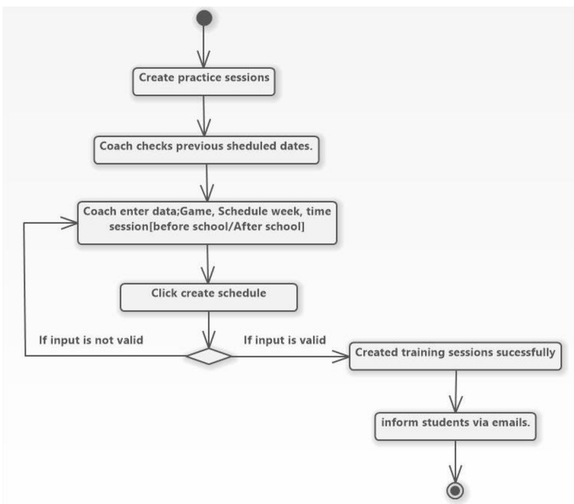


Figure 3.8: Activity Diagram for the Create training sessions.

3.9. DATABASE DESIGN

A database is an organized collection of data. The data are typically organized to model relevant aspects of reality in a way that supports processes requiring this information. Database design is the process of producing a detailed data model of a database. This logical data model contains all the needed logical and physical design choices and physical storage parameters needed to generate a design in a data definition language, which can then be used to create a database. A fully attributed data model contains detailed attributes for each entity.

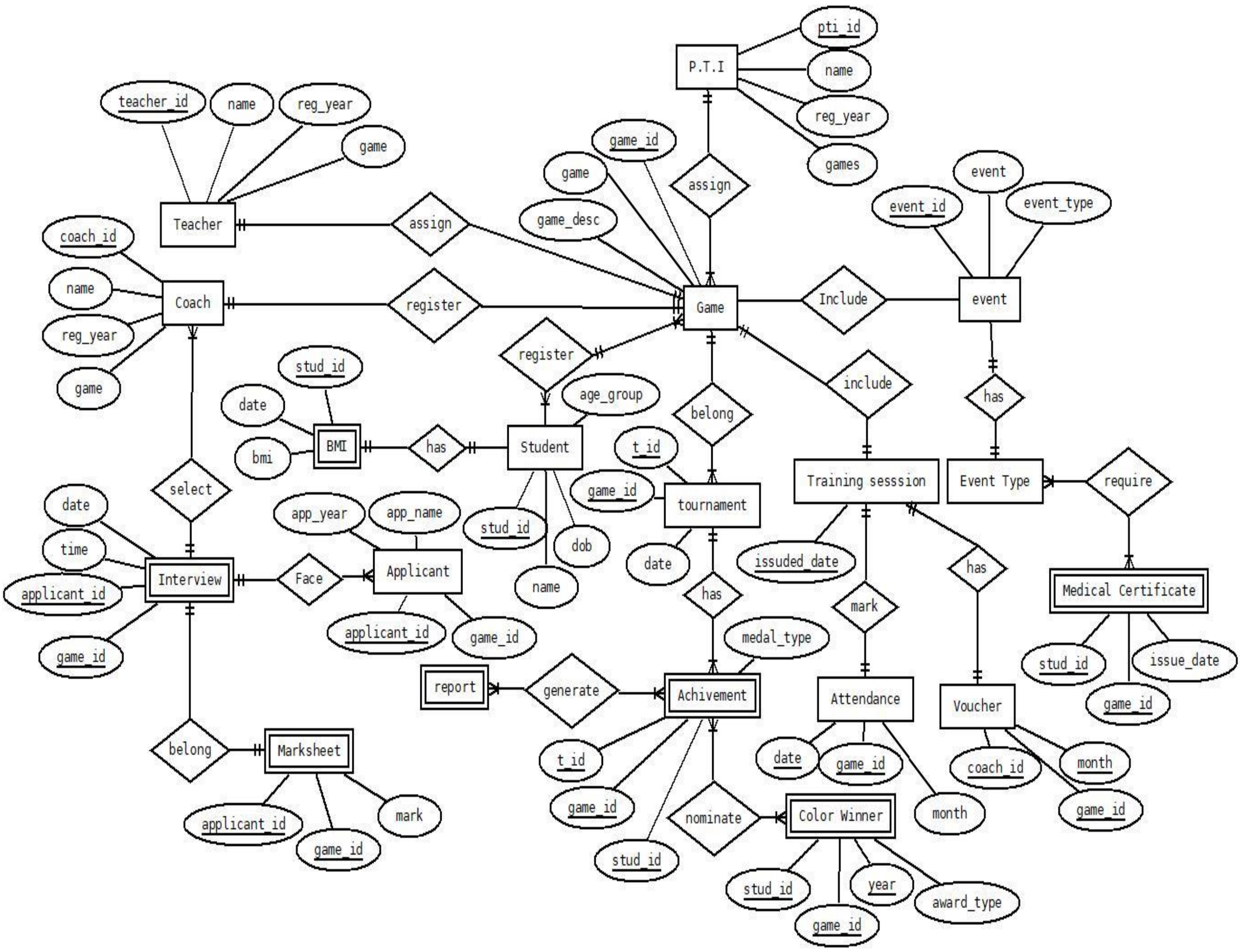


Figure 3.9 – ER Diagram.

3.10 USER INTERFACE DESIGN

The user and the system are interacting by using user interfaces. User is allowed to interact with the system without bothering about the back end logics. A bootstrap template is used to designing the attractive interfaces in the system.

Following main user interface designing practices will be followed while designing the interfaces.

- Keep the interface simple
- Select fonts and font sizes which is clear and readable
- Use attractive color range to make user attraction
- Set navigations with main interfaces

3.11 MAIN INTERFACES

3.11.1 HOME PAGE

Figure 3.10 illustrates the home page of the system. This interface allows main users, PTI and Teachers to register and external users to apply as a Coach.

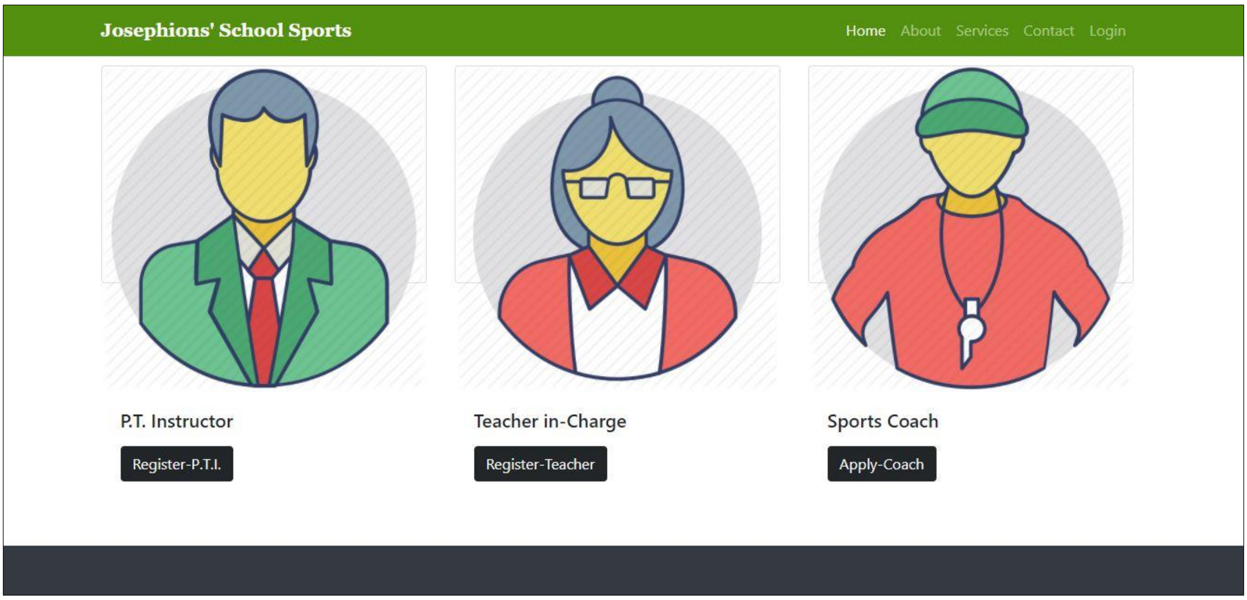


Figure 3.10 Home Page of the system

3.11.2 LOGIN PAGE

Figure 3.11 illustrates the Login screen of the system. This interface allows the user to log into the system. Main actors can create own User Name and Passwords.

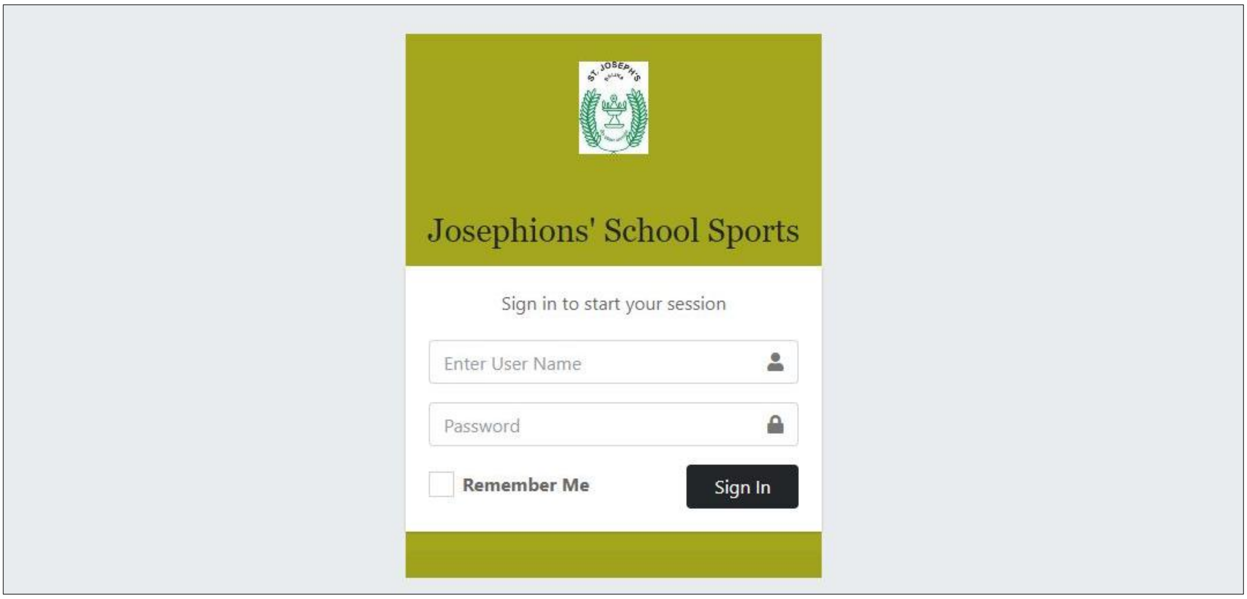


Figure 3.11 Login screen of the system

3.11.2 REGISTRATION FORMS OF MAIN SYSTEM USERS.

Figure 3.12 and 3.13 illustrate two steps of data entry form of the system. This form is used to register school PTI to the system.

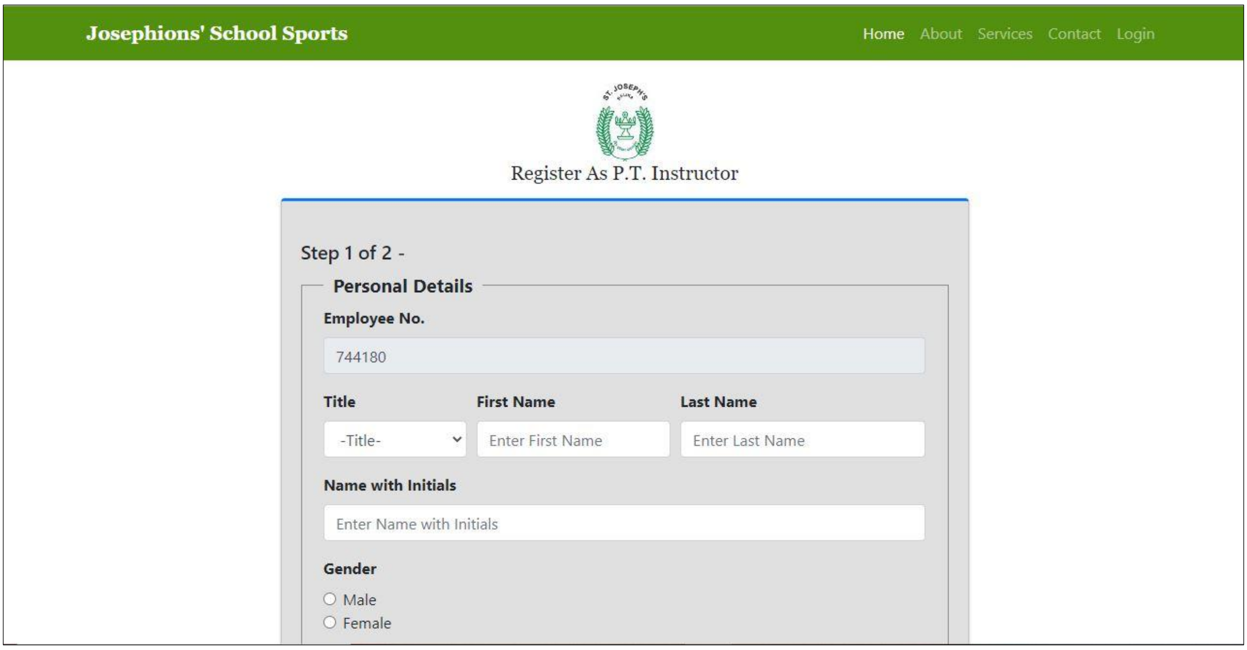


Figure 3.12 User Registration form of the system

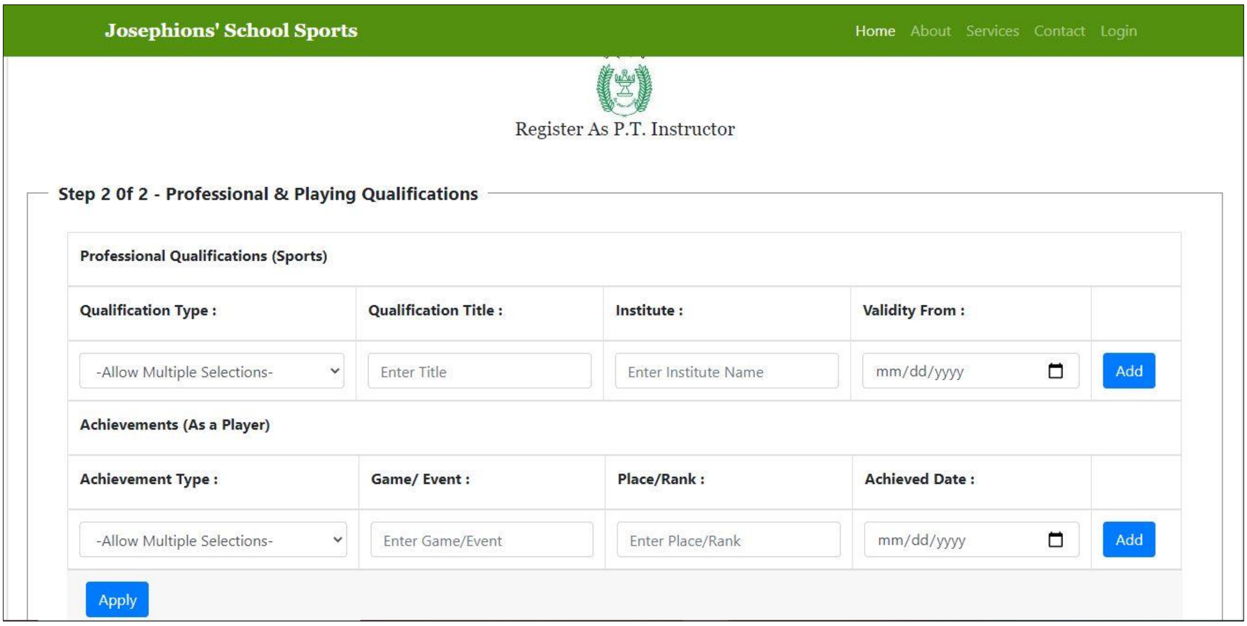


Figure 3.13 User Registration form of the system

3.11.3 REGISTRATION VERIFICATION MESSAGES.

Figure 3.14 is shown the registration verification messages.

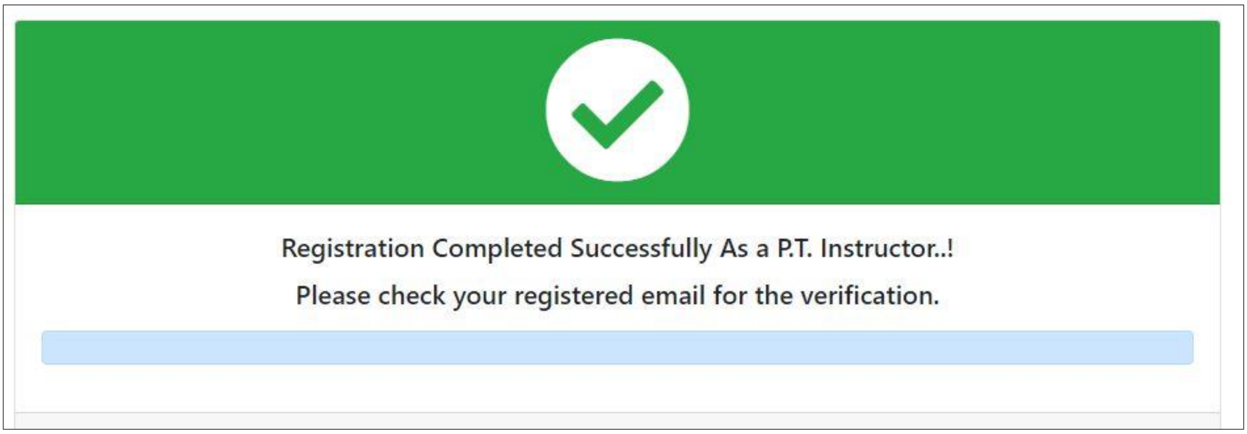


Figure 3.14 Registration Verification Message.

3.11.4 DASH BOARD

Figure 3.15 illustrates the principal's dash board of the system. It contains the import facts for the management authority of the school.

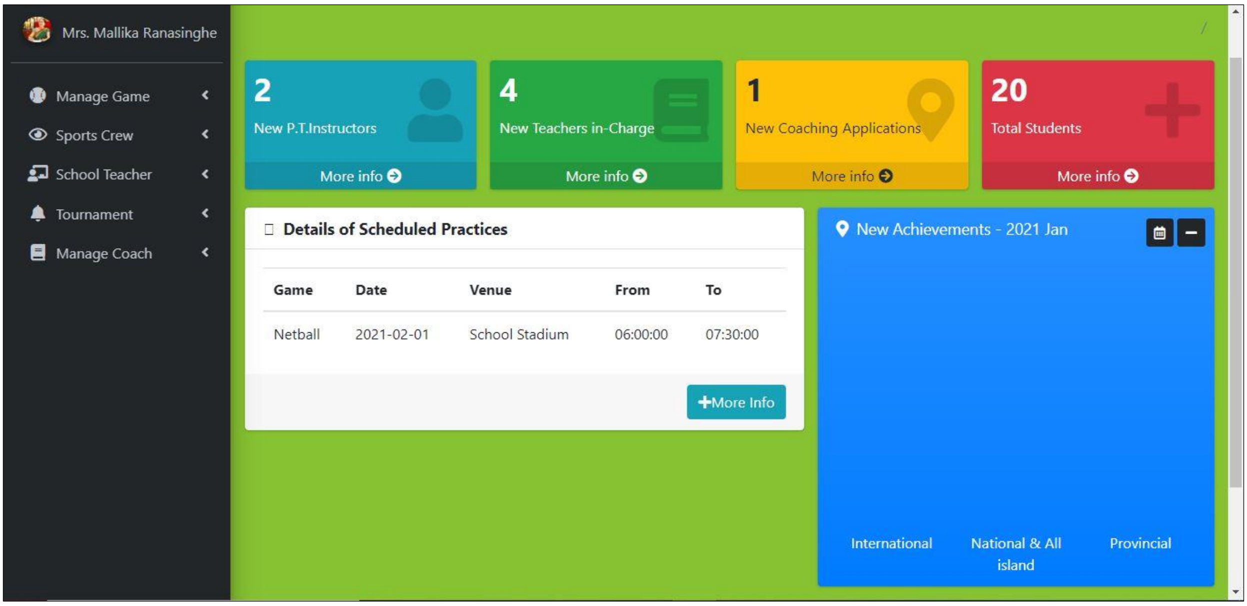


Figure 3.15 Principal Dash Board of the System

3.11.5 ASSIGN GAMES FOR SPORTS STAFF

Figure 3.16 illustrates the form of assigning games for the registered PTI. Important tips that are helpful to this function are displayed on this screen.

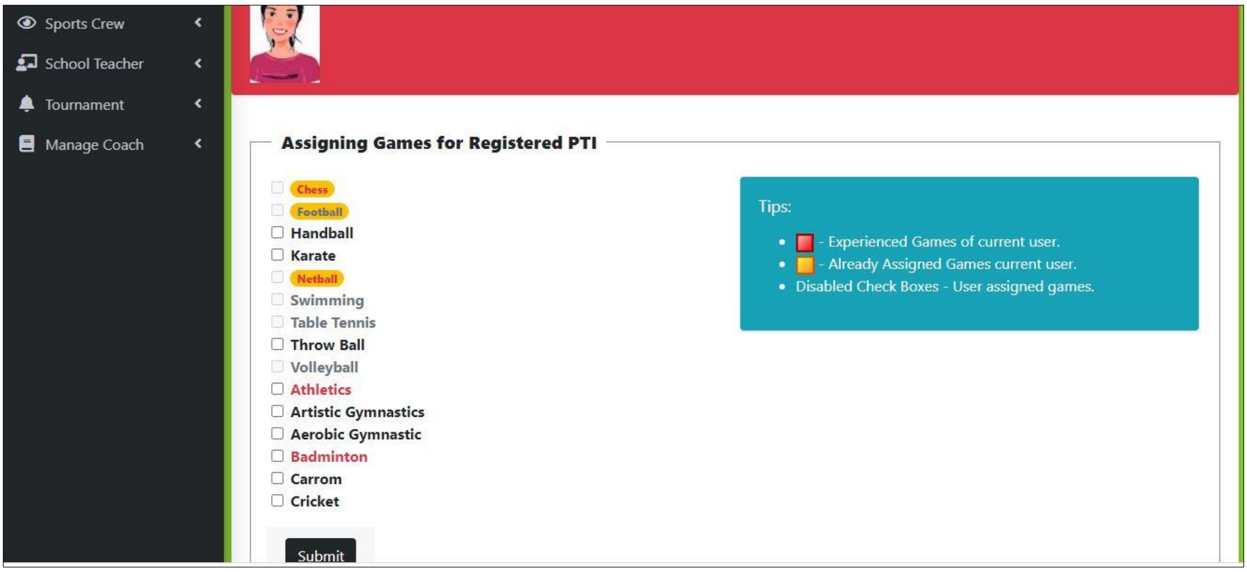


Figure 3.16 Games Assigning form of the System

3.11.6 ATHLETIC SKILL TEST

Figure 3.17 is shown the interface of athletics skill test. It allows the coaches to identify athlete’s skill type.

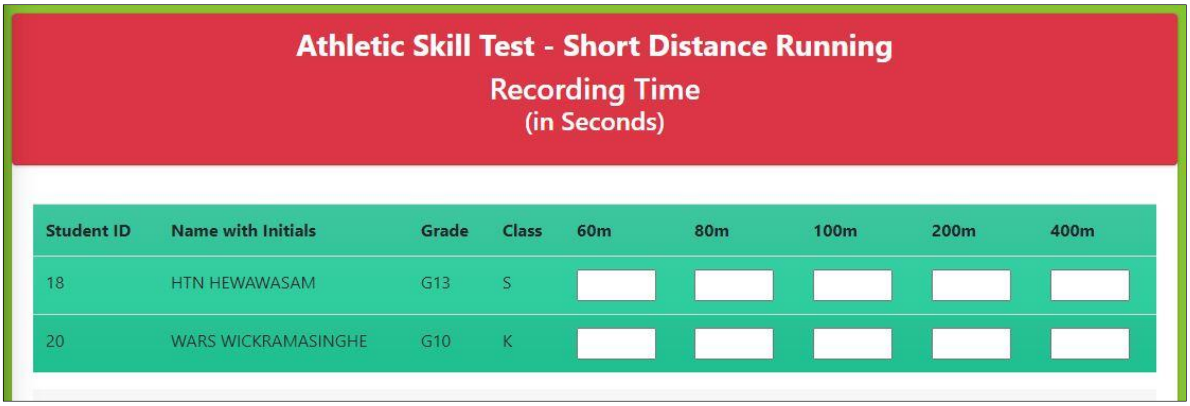


Figure 3.17 Athletic Skill Test

3.11.7 ATTENDANCE MARKING

Figure 3.18 is shown the attendance marking of the students for each scheduled practice.

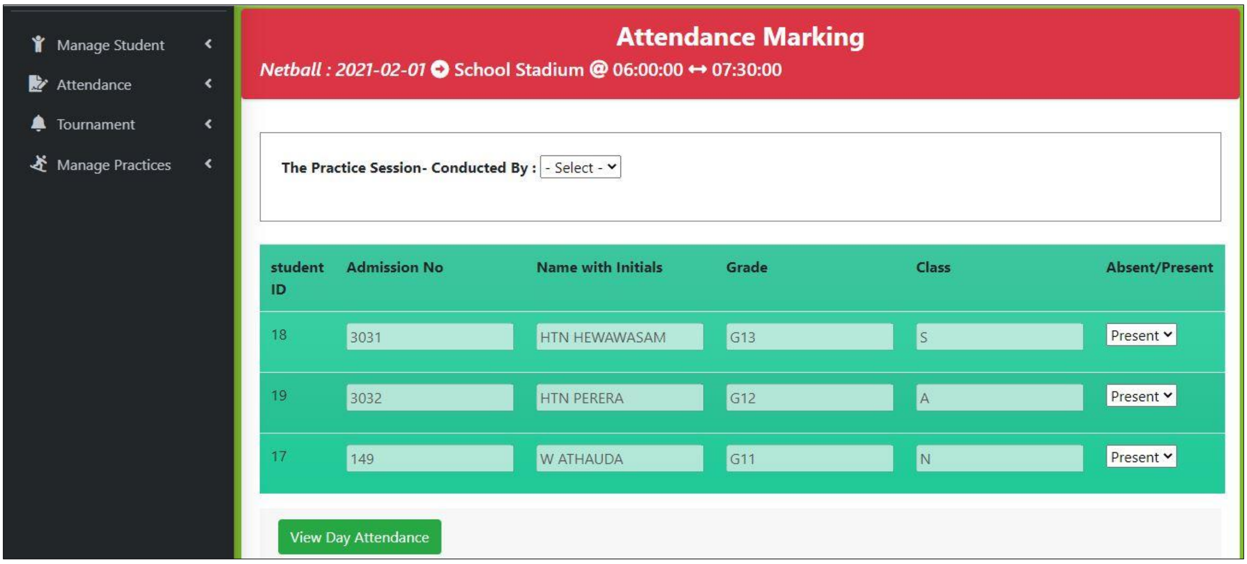


Figure 3.18 Attendance Marking

3.11.8 BMI CALCULATING

Figure 3.19 is shown the latest calculated BMI of the students and text boxes for insert measurements for next calculation.

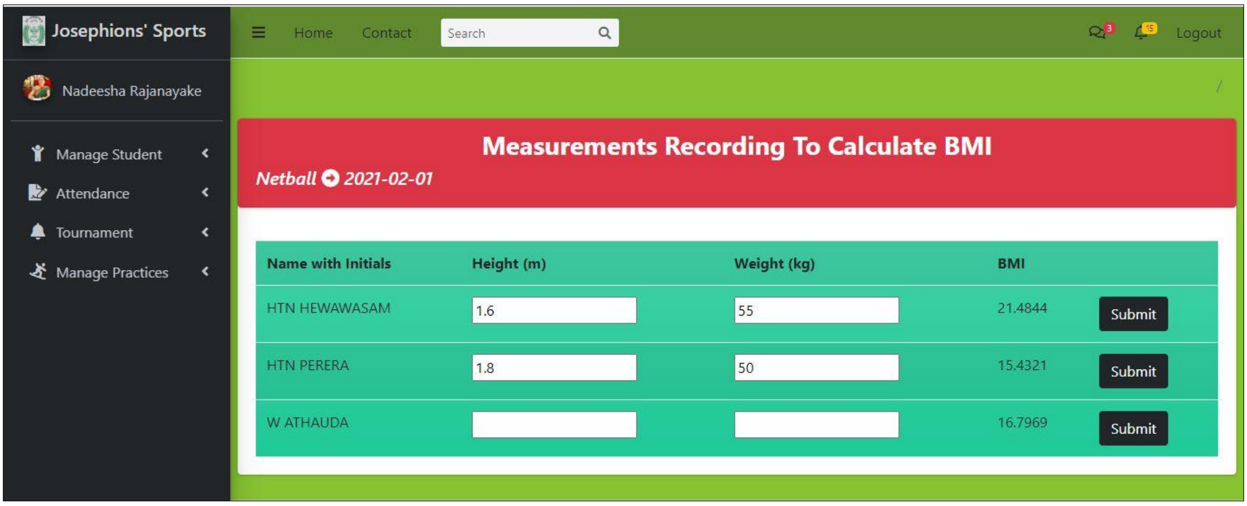


Figure 3.19 Calculate BMI

3.11.9 SELECTION STUDENTS FOR TOURNAMENTS.

3.11.10 RECORD ACHIEVEMENTS AND DECIDE STUDENT COLOR AWARD TYPE.

CHAPTER 4: IMPLEMENTAION

4.1 INTRODUCTION

Implementation is the process of construction of the new system according to what the design stage planed by using appropriate tools and techniques. System designing and the implementation process transforms the system specification in to an executable program which can use for day-to-day business or organizational operations.

This system has been developed as separate modules and finally integrated all of them as a complete system. This chapter mainly discussed about the implementation environment and the tools and techniques has been used to develop the system. As well as included major code segments have been used along with comments which can be useful for further improvements.

4.1 IMPLEMENTAION ENVIRONMENT

The following table 4.1 showing the hardware and software components were used in the implementation environment. It was concerned to select free and open-source software when selecting the development software.

4.2.1 HARDWARE AND SOFTWARE ENVIRONMENT

Hardware	Software/ Web Application hosting
• AMD Ryzen TM 3 2200U • 2.5 GHz • 4GB RAM • 1 TB Hard Disk	• Microsoft Windows 10 • XAMPP v3.2.2 o PHP 7.2.30 o MySQL client version 5.6 o Apache 2.4.43 o Php MyAdmin 5.0.2

Table 4.1 Implementation Environment

4.2.2 DEVELOPMENT TOOLS AND TECHNOLOGIES

When developing the system, the following tools and technologies were used.

- NetBeans IDE 11.2 – for coding of the system
- Software Ideas Modeler – for diagram designing

- PHP (Hypertext Pre-Processor) – This is the server-side object-oriented scripting language, which was mainly used to develop main system and its logics.
- HTML - was used to build the base Interfaces of the software
- CSS - was used to format the plain HTML interfaces more attractively
- JavaScript - was used to code all the client-side validations
- AJAX - This technique is used to make the interfaces more interactive by checking for any field values from the server without reloading the page.

4.3 REUSED MODULES

Bootstrap template -

"Bootstrap is a free and open-source CSS framework directed at responsive, mobile-first front-end web development. It contains CSS and (optionally) JavaScript-based design templates for typography, forms, buttons, navigation, and other interface components." [7]

Phpmailer -

PHPMailer is one of the most popular open-source PHP libraries to send emails.

4.4 ARCHITECTURAL OVERVIEW

This system is developed based on the **Model-View-Controller (MVC)** software architecture.

The **Model-View-Controller (MVC)** is an architectural pattern that separates an application into three main logical components: the model, the view, and the controller. Each of these components are built to handle specific development aspects of an application.

MVC is one of the most frequently used industry-standard web development framework to create scalable and extensible projects. The lowest level of the pattern which is responsible for maintaining data. [8]

VIEW - This is responsible for displaying all or a portion of the data to the user. Presentation of data in a particular format, triggered by a controller's decision to present the data. They are script based templating systems like JSP, ASP, PHP and very easy to integrate with AJAX technology.

CONTROLLERS - Software Code that controls the interactions between the Model and View.

The controller receives the input, it validates the input and then performs the operation that modifies the state of the data model.

The following Figure 4.1 is described the process of the MVC architecture.

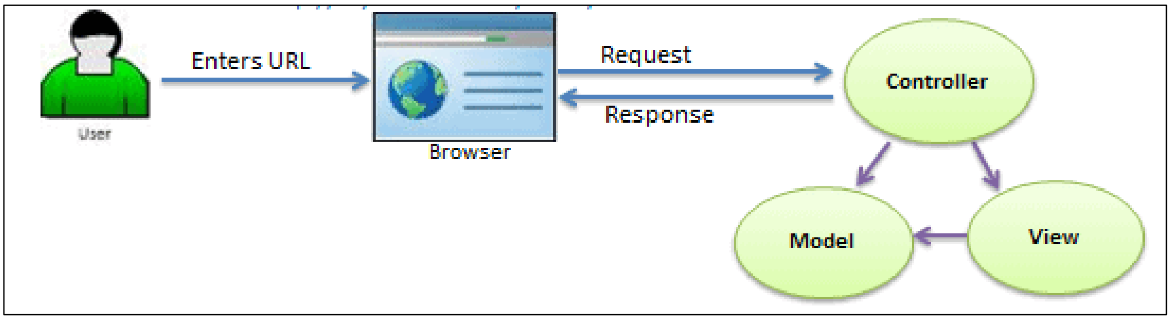


Figure 4.1 MVC Software Architecture

4.5 MAJOR CODE SEGMENTS

The major code segments of the system have been briefly described as following with functionalities and comments of code lines.

DATABASE CONNECTION PAGE

Database connection is an extremely important step because when some data is added, deleted, updated in the system, the related database always connects with the system. The system is connected to the database by this code.

```
<?php

//Set database connection attributes
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "db_ssmgtsys";
//create new database object
$conn = new mysqli($servername, $username, $password, $dbname);
//Check connection
if ($conn->connect_error) {
    die("Connection failed" . $conn->connect_error);
}
?>
```

LOGIN HANDLING PAGE

The code segment given below gets the data from the users table and checks that the user exists according to the user input. If exists check the equality of database's password with input password. If they matched, check the login type and refer to the relevant user's dashboard.

```php
<?php
session_start();
include 'helper.php';
include 'conn.php';
//Check the request method
if ($_SERVER['REQUEST_METHOD'] == "POST") {
    //Define variables as empty
    $user_name = $password = null;
    $e = array();
    //Clean inputs
    $user_name = clean_input($_POST['user_name']);
    $password = clean_input($_POST['password']);
    //Empty validation
    if (empty($user_name)) {
        $e['user_name'] = "User name should not be empty";
    }
    if (empty($password)) {
        $e['password'] = "Password should not be empty";
    }
    //If No errors query will be executed..
    if (empty($e)) {
        $sql = "SELECT * FROM tb_user WHERE user_name='$user_name' and password='" . sha1($password) . "'";
        $result = $conn->query($sql);
        if ($result->num_rows == 1) {
            while ($row = $result->fetch_assoc()) {
                $_SESSION['user_id'] = $row['user_id'];
                $_SESSION['first_name'] = $row['first_name'];
                $_SESSION['last_name'] = $row['last_name'];
                $_SESSION['role'] = $row['role'];
            }
            header('Location:dashboard.php');
```

INSERT DATA INTO TABLES

```php
<?php include '../header.php'; ?>
<?php include '../nav.php'; ?>
<?php
if ($_SERVER['REQUEST_METHOD'] == "POST") {
//Begin of Variable Declaration
    $empno = $nic = $appoint_subject = $appoint_date = $section = null;
//Begin of Clean Inputs
    $e = array();
    $empno = clean_input($_POST['empno']);
    $nic = clean_input($_POST['nic']);
    $appoint_subject = clean_input($_POST['appoint_subject']);
    $appoint_date = clean_input($_POST['appoint_date']);
    if (isset($_POST['section'])) {
        $section = $_POST['section'];
    }
// Begin of empty validation
    if (empty($empno)) {
        $e['empno'] = "Emp No should not be empty...";
    }
    if (empty($nic)) {
        $e['nic'] = "NIC should not be empty...";
    }
    if (empty($section)) {
        $e['section'] = "Section should not be empty...";
    }
    if (empty($appoint_subject)) {
        $e['appoint_subject'] = "Appoint Subject should not be empty...";
    }
```

```php
    }
//Begin of Advanced Validation
    //Limit only to input letters and numbers.
    if (!empty($empno)) {
        if (!preg_match("/^[0-9]*$/", $empno)) {
            $e['empno'] = "Only numaric values are allowed...";
        }
    }
    //Checks whether empno is Exist.
    if (!empty($empno)) {
        $sql = "SELECT * FROM tb_master_teacher WHERE empno='$empno'";
        $result = $conn->query($sql);
        if ($result->num_rows > 0) {
            $e['empno'] = "This Empno already exsist.";
        }
    }
    //Check whether NIC is in a valid format.
    if (!empty($nic)) {
        if (!preg_match("/^([0-9]{9}[x|X|v|V]|[0-9]{12})$/", $nic)) {
            $e['nic'] = "Enter a valid NID format";
        }
    }
//Insert records into table, if no validation errors.
    if (empty($e)) {
        $str = "INSERT INTO `tb_master_teacher`(`empno`,`nic`,`appoint_subject`,`appoint_date`,`section`) VALUES ('$empno'

        $r = $conn->query($str);
        if ($r === TRUE) {
            echo "<div class='alert alert-success' id='msg'>School Teacher Successfully Added.</div>";
```

UPDATE TABLE

```php
<?php include '../header.php'; ?>
<?php include '../nav.php'; ?>
<?php
$empno = $title = $name_initials = $first_name = $last_name = $gender = $nic = $address = $mobile_num = $home_num = $emai

if ($_SERVER['REQUEST_METHOD'] == "POST" && @$_POST['operate'] == "edit") {
    $user_id = $_POST['user_id'];
    $sql = "SELECT * FROM tb_teachers WHERE user_id='$user_id'";
    $result = $conn->query($sql);

    if ($result->num_rows > 0) {
        while ($row = $result->fetch_assoc()) {

            $empno = $row['empno'];
            $title = $row['title'];
            $first_name = $row['first_name'];
            $last_name = $row['last_name'];
            $gender = $row['gender'];
            $nic = $row['nic'];
            $address = $row['address'];
            $mobile_num = $row['mobile_num'];
            $home_num = $row['home_num'];
            $email = $row['email'];
            $profile_image = $row['profile_image'];
        }
    }
}
```

```
if ($_SERVER['REQUEST_METHOD'] == "POST" && @$_POST['operate'] == 'update') {

    $e = array();
//Begin of clean inputs
    $user_id = $_POST['user_id'];
    $empno = clean_input($_POST['empno']);
    if (isset($_POST['title'])) {
        $title = $_POST['title'];
    }
    $name_initials = clean_input($_POST['name_initials']);
    $first_name = clean_input($_POST['first_name']);
    $last_name = clean_input($_POST['last_name']);
    if (isset($_POST['gender'])) {
        $gender = $_POST['gender'];
    }

    $nic = clean_input($_POST['nic']);
    $address = clean_input($_POST['address']);
    $mobile_num = clean_input($_POST['mobile_num']);
    $home_num = clean_input($_POST['home_num']);
    $email = clean_input($_POST['email']);
//Begin of Empty Validation
    if (empty($empno)) {
        $e['empno'] = "Emp No should not be empty...";
    }
    if (empty($title)) {
        $e['title'] = "Title should not be empty...";
    }
    if (empty($name_initials)) {
```

```
//Insert records into table, if no validation errors.
    if (empty($e)) {
        $str = "UPDATE `tb_games` SET `game`='$game',`game_desc`='$game_desc',`game_logo`='$game_logo' WHERE game_id='$gar

        $r = $conn->query($str);
        if ($r === TRUE) {
            echo "Data Inserted Successfully...";
        }
    }
}
?>
```

DELETE DATA FROM TABLE

```
<?php
include "../conn.php";
$user_id = $_POST['user_id'];
$sql = "DELETE FROM tb_teachers WHERE user_id='$user_id'";
$conn->query($sql);
//After deleting record return to pravious page.
header('Location:view_reg_teacher.php');
?>
```

SEARCH DETAILS ACCORDING TO GIVEN CRITERIA

```
<?php include '../header.php'; ?>
<?php include '../nav.php'; ?>
<?php
if ($_SERVER['REQUEST_METHOD'] == "POST" && @$_POST['operate'] == "search") {

    $game_id = $_POST['s_game'];
    $sql = "SELECT * FROM tb_applied_coach LEFT JOIN tb_title ON tb_applied_coach.application_id=tb_title.title_id "
            . "LEFT JOIN tb_games ON tb_applied_coach.applied_game=tb_games.game_id "
            . "WHERE tb_applied_coach.applied_game='$game_id'";
    $result = $conn->query($sql);
} else {
    $sql = "SELECT * FROM tb_applied_coach "
            . "LEFT JOIN tb_title ON tb_applied_coach.application_id=tb_title.title_id "
            . "LEFT JOIN tb_games ON tb_applied_coach.applied_game=tb_games.game_id";
    $result = $conn->query($sql);
}
?>
```

4.7 REUSABLE COMPONENTS

The following re-usable components have been used when implementing the system to identify uniquely, add more attractiveness and to maximize the efficiency of the system.

- **Top Navigation Bar**
  In the system, top navigation bar codes have been used when designing system interfaces
- **Footer**
  To preserve system uniqueness, footer codes have been used when designing system interface.
- **System Messages**
  When login to the system, when save some records and when delete a record system will provide a relevant message. The user will be able to identify the relevant error.
- **Combo Box Search**
  Combo box search has been used to show data which are stored in database, and search data according to specific key word

4.7 DATA VALIDATION

Data validation is a most important and critical of the implementation process. Because users can enter unwanted values to the system, so that accuracy of the data can be loosed. The main objective of the data validation is ensuring the input value is in the correct format.

Following code snippets of data validation are shown in Figure 4.2

```php
//Begin of Advanced Validation
    //Checks whether empno is Exist.
    if (!empty($empno)) {
        $sql = "SELECT * FROM tb_teachers WHERE empno='$empno'";
        $result = $conn->query($sql);
        if ($result->num_rows > 0) {
            $e['empno'] = "This Empno already exsist.";
        }
    }
    //Checks if name only contains letters,whitespaces & dots.
    if (!empty($name_initials)) {
        if (!preg_match("/^[a-zA-Z .]*$/", $name_initials)) {
            $e['name_initials'] = "Only letters, white spaces and dots allowed";
        }
    }
    //Converts 'Name with Initials' into UPPERCASE.
    if (!empty($name_initials)) {
        $name_initials = strtoupper($name_initials);
    }
    //Checks whether 'First name' & 'Last name' only contains letters.
    if (!empty($first_name)) {
        if (!preg_match("/^[a-zA-Z]*$/", $first_name)) {
            $e['first_name'] = "Only letters are allowed..";
        }
    }
    if (!empty($last_name)) {
        if (!preg_match("/^[a-zA-Z]*$/", $last_name)) {
            $e['last_name'] = "Only letters are allowed..";
        }
    }
```

```php
    //Password Validation
    if (!empty($password)) {
        if (strlen($_POST["password"]) <= 6) {
            $e['password'] = "Your Password Must Contain At Least 6 Characters!";
        } elseif (!preg_match("#[0-9]+#", $password)) {
            $e['password'] = "Your Password Must Contain At Least 1 Number!";
        } elseif (!preg_match("#[A-Z]+#", $password)) {
            $e['password'] = "Your Password Must Contain At Least 1 Capital Letter!";
        } elseif (!preg_match("#[a-z]+#", $password)) {
            $e['password'] = "Your Password Must Contain At Least 1 Lowercase Letter!";
        } elseif (!preg_match("#[\W]+#", $password)) {
            $e['password'] = "Your Password Must Contain At Least 1 Special Character!";
        }
    }
    //Compare two passwords
    if ($_POST["password"] === $_POST["confirm_password"]) {
        // success
    } else {
        $e["confirm_password"] = "Please make sure your passwords match.";
    }
    //Check whether NIC is in a valid format.
    if (!empty($nic)) {
        if (!preg_match("/^([0-9]{9}[x|X|v|V]|[0-9]{12})$/", $nic)) {
            $e['nic'] = "Enter a valid NID format";
        }
    }
    // compare nic with gender
    if (!empty($nic)) {
        if (strlen($nic) == 10) {
            $gen = substr($nic, 2, 3);
```

Figure 4.2 Data Validation

CHAPTER 5 – EVALUATION

5.1 INTRODUCTION

This chapter is included the testing phase of the system. Testing is the process of evaluating a system or its component with the intent to find whether it satisfies the specified requirements or not. Further testing is executing a system in order to identify any gaps, errors or missing requirements in contrary to the actual desire or requirements

5.2 TESTING PROCEDURE

The testing is not a separate process. It begins with the implementation process. The system should be tested for the errors while coding the system.

Software testing is very important because of the following reasons:

1. Software testing is really required to point out the defects and errors that were made during the development phases.
2. It‟s essential since it makes sure of the users' reliability and their satisfaction in the application.
3. It is very important to ensure the reliability and accuracy of information. Reliable and accuracy information helps in gaining system users' confidence.
4. Testing is required for an effective performance of software application.
5. It's important to ensure that the application should not result into any failures because it can be very expensive in the future or in the later stages of the development.

The various levels of testing are;

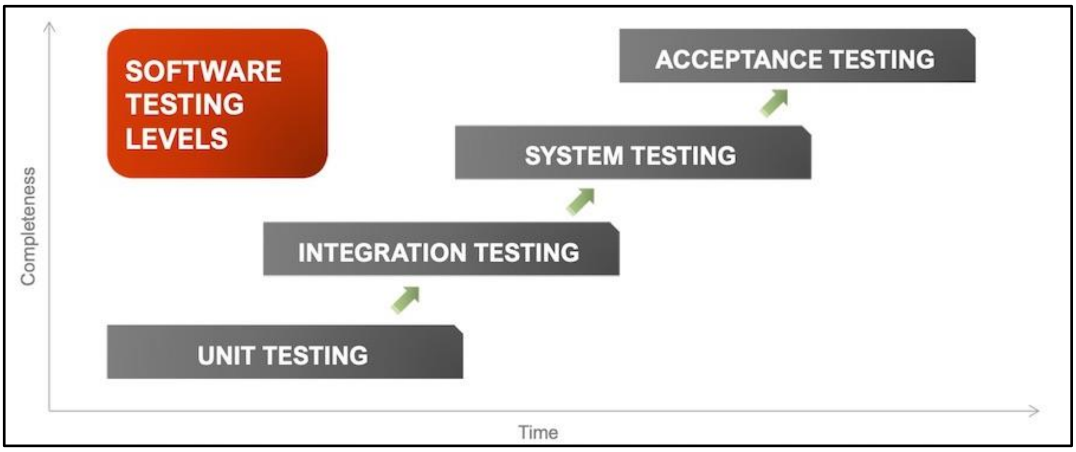


Figure 5.1 Levels of System Testing

1. **Unit testing** - A level of the software testing process where individual units of a software are tested. The purpose is to validate that each unit of the software performs as designed.

   This can be black box testing or White box testing. [9]

Black Box Testing is a software testing method in which the functionalities of software applications are tested without having knowledge of internal code structure, implementation details and internal paths. [10] Tester is mainly concerned with the validation of output rather than how the output is produced.

White Box Testing is software testing technique in which internal structure, design and coding of software are tested to verify flow of input-output and to improve design, usability and security. [11]

Programming knowledge and implementation knowledge (internal structure and working) is required in White Box testing, which is not necessary in Black Box testing.

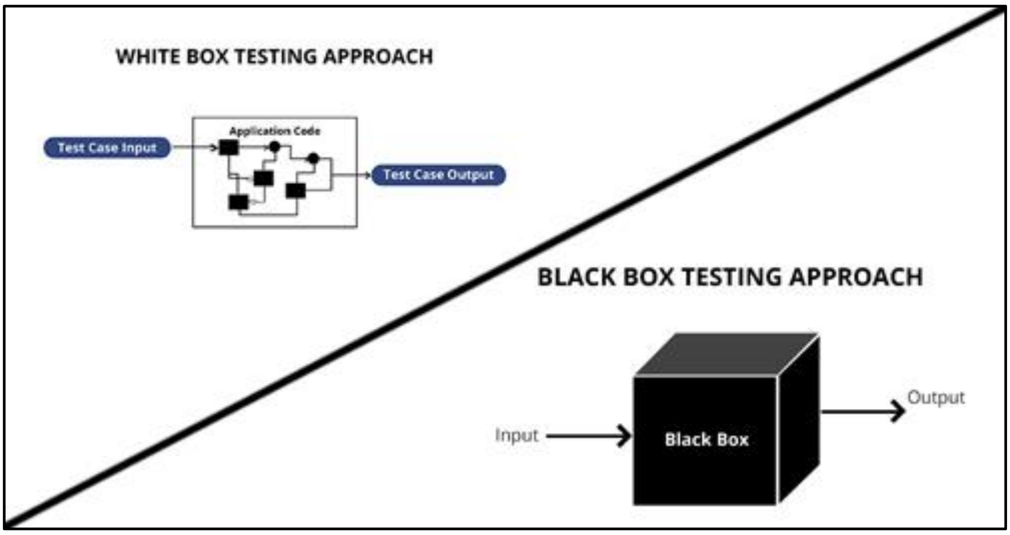


Figure 5.2 Comparison among Black-Box and White-Box tests

2. **Integration Testing**: A level of the software testing process where individual units are combined and tested as a group. The purpose of this level of testing is to expose faults in the interaction between integrated units.
3. **System testing:** A level of the software testing process where a complete, integrated system is tested. The purpose of this test is to evaluate the system's compliance with the specified requirements.

4. **Acceptance testing:** A level of the software testing process where a system is tested for acceptability. The purpose of this test is to evaluate the system's compliance with the business requirements and assess whether it is acceptable for delivery.

5.2 TEST PLAN AND TEST CASES

In the testing procedure, most important part is creating test cases. Once creating the test strategy, test case creation is done.

Generally, test case consists description of test case, steps to test, expected output and priority. In order to mitigate the difficulty of the system, system has divided into modules. Test cases were written for each module.

5.2.1 TEST CASE FOR USER AUTHENTICATION

ID	Test Description	Steps to Test	Expected Result	Priority
01	Login in to the system (Negative)	Submit login form with empty fields	Display an error message like "User name should not be empty" and "Password should not be empty"	High
02	Login in to the system (Negative)	Enter an invalid user name or password	Display an error message like "Invalid User name or Password ."	High
03	Login in to the system (Positive)	Enter a valid user name and password	Redirect to the users' dashboard	High
04	Display user name after successfully login (Positive)	Login to the system with valid user name	Display logged username on the top status bar	Medium
05	Privilege module access according to the user type (Positive)	Login to the system as Principal	Display Principal's menu only	High
06	Logout form the system	Click on the given logout button	The user can log out from the system and redirect to the login page	High

Table 5.1: Test cases of user Authentication

The user login module is done by the registered users of the developed system with the user name and password. The Table 5.1 is shown the test cases of login function.

5.2.2 TEST CASE FOR USER REGISTRATION

ID	Test Description	Steps to Test	Expected Result	Priority
01	Confirm user as School Teacher. (Negative)	Enter invalid Emp no or NIC Number.	Show the message “Sorry, you can’t go forward, because you are not in School Teacher Database”	High
02	Check whether user already registered. (Negative)	Enter already registered Emp. number.	Show the message “This user already registered”	High
03	Limit characters in ‘name with initials’ (Negative)	Enter name with comma.	Show the message “only letters, white spaces and dots are allowed”	Medium
04	Check user name existence. (Negative)	Enter existing user name.	Show the message “This user name already exists.”	High
05	Empty Validation. (Negative)	Not fill the fields of the form.	Show the massage under each field such as “DOB should not be empty”	High
06	Check for the valid NIC No. (Negative)	Enter incorrect NIC number.	Show the massage “Enter a valid NIC format.”	High
07	Check for the valid telephone number. (Negative)	Enter incorrect telephone number.	Show the massage “Invalid phone number”	High
08	Check the accuracy of email address. (Negative)	Enter incorrect email address.	Show the massage “Invalid e-mail format”	High
09	Check availability of the email address. (Negative)	Enter existing email address.	Show the massage “ This Email already exist and please add another email address.”	High
10	Check availability of uploaded image file (Negative)	Upload image file with existing file name.	Show the message "Sorry file already exists."	High
11	Limit the image file size. (Negative)	Upload image size within more than 500000 bits.	Show the message “sorry, your file is too large.”	High

12	Check uploaded file type. (Negative)	Upload non image file type	Show the message "Sorry, only jpg, png, jpeg, png and gif files are allowed."	High
13	Confirm password. (Negative)	Enter unmatched confirm password.	Show the message “Please make sure your passwords match. .”	High
14	Check password length. (Negative)	Enter password less than 6 characters.	Show the message “ Your Password Must Contain At Least 6 Characters!”	High
15	Check password content. (Negative)	Enter password without Uppercase.	Message “Your Password Must Contain At Least one capital letter”	High
16	Check password content. (Negative)	Enter password without lowercase.	Message “Your Password Must Contain At Least one simple letter”	High
17	Check password content. (Negative)	Enter password without special character.	Message “Your Password Must Contain At Least one special character”	High

Table 5.2.3 : TEST CASES FOR SCHEDULE COACH INTERVIEW

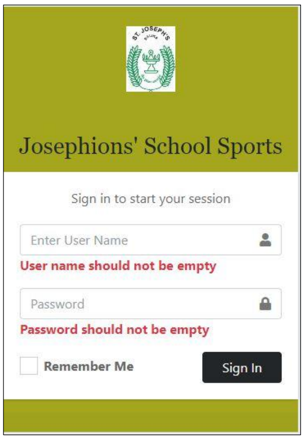
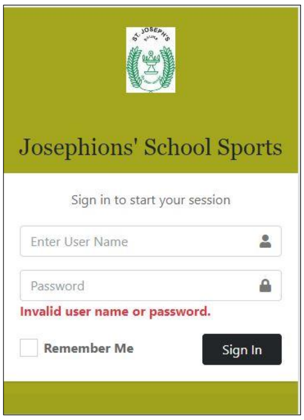
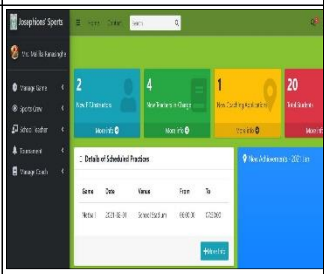
ID	Test Description	Steps to Test	Expected Result	Priority
01	Schedule Coach Interview (Negative)	Check date is a past date.	Show the message “Past date..! Please enter future date”	High
02	Schedule Coach Interview (Negative)	Check entered date and time-availability of scheduling.	Show the message “This date & Time is already reserved.”	High

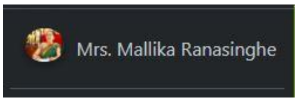
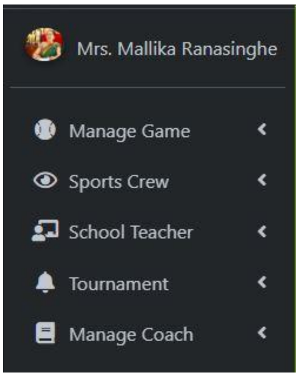
Table 5.3: Test cases of schedule Coach Interview

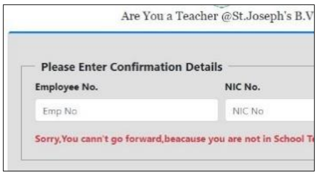
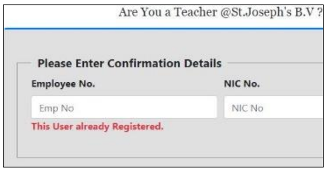
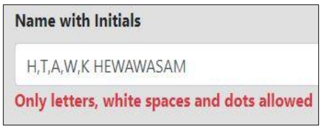
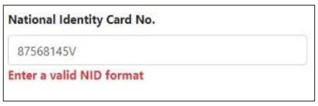
5.3 TEST RESULT

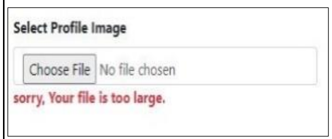
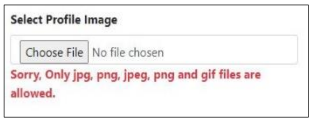
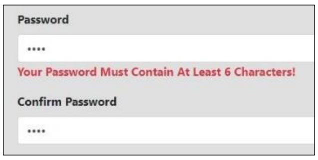
Evidence should be produced to show that all aspects of the system have been tested and specification has been met. A common set of error messages were reused in relevant places to reduce confusion by given different error messages for same functionalities.

Test cases are shown as below in table 5.1;

ID	Step of the Test	Expected Result	Actual Result	Status
USER AUTHENTICATION				
01	Submit login form with empty fields	Display an error message like “Please enter username or password”		Pass
02	Enter an invalid user name or password (Negative)	Display an error message like “Invalid user name or password.”		Pass
03	Enter a valid user name and password (Positive)	Redirect to the dashboard (home page)		Pass

04	Login to the system with valid user name	Login to the system with valid user name		Pass
05	Login to the system as Principal	Display Principal's menu only		Pass
06	Logout form the system	Click on the given logout button		Pass

ID	Step of the Test	Expected Result	Actual Result	Status
USER REGISTRATION				
07	Enter invalid Emp no or NIC Number to confirm.	Show the message “Sorry, you can’t go forward, because you are not in School Teacher Database”		Pass
08	Enter already registered Emp number.	Show the message “This user already registered”		Pass
09	Enter name with comma.	Show the message “only letters, white spaces and dots are allowed”		Pass
10	Enter existing user name.	Show the message “This user name already exists.”		Pass
11	Not fill the fields of the form.	Show the massage under each field such as “DOB should not be empty”		Pass
12	Enter incorrect NIC number.	Show the massage “Enter a valid NIC format.”		Pass
13	Enter incorrect telephone number.	Show the massage “Invalid phone number”		Pass
14	Enter incorrect email address.	Show the massage “Invalid e-mail format”		Pass

15	Enter existing email address.	Show the massage " This Email already exist and please add another email address."		Pass
16	Upload image file with existing file name.	Show the message "Sorry file already exists."		Pass
17	Upload image size within more than 500000 bits.	Upload image size within more than 500000 bits.		Pass
18	Upload non image file type	Show the message "Sorry, only jpg, png, jpeg, png and gif files are allowed."		Pass
19	Enter unmatched confirm password.	Show the massage "Please make sure your passwords match. ."		Pass
20	Enter password less than 6 characters.	Show the message " Your Password Must Contain At Least 6 Characters!"		Pass
21	Enter password without Uppercase.	Message "Your Password Must Contain At Least one Capital letter"		Pass
22	Enter password without lowercase.	Message "Your Password Must Contain At Least one Simple Letter"		Pass
23	Enter password without special character.	Message "Your Password Must Contain At Least one Special Character"		Pass

ID	Step of the Test	Expected Result	Actual Result	Status
SCHEDULE COACH INTERVIEW				
24	Check date is a past date.	Show the message “Past date..! Please enter future date”		**Pass**
25	Check entered date and time-availability of scheduling.	Show the message “This date & Time is already reserved.”		**Pass**

5.4 USER ACCEPTANCE TEST

User Acceptance Testing (Beta testing) is the last phase of software development in which the software is tested in the real environment by the intended audience. It gives users the chance to interact with the software and ensuring that the application behaves exactly as expected and detect faults and defects.

System Users:

Principal, PTI, Teacher in-charge, Sports Coach, Student

The questionnaire which is one of the testing tools was used to increase the efficiency and accuracy of the test results. This tool was helped to trace the actual ideas and results from the users of the developed system.

Feedback gathered from the questionnaire and clients’ comments were recorded to make few modifications to the system. Overall feedback was positive and satisfying.

The user evaluation form can be found in Appendix E

Following Figure 5.1 is shown the summery of the user feedback.

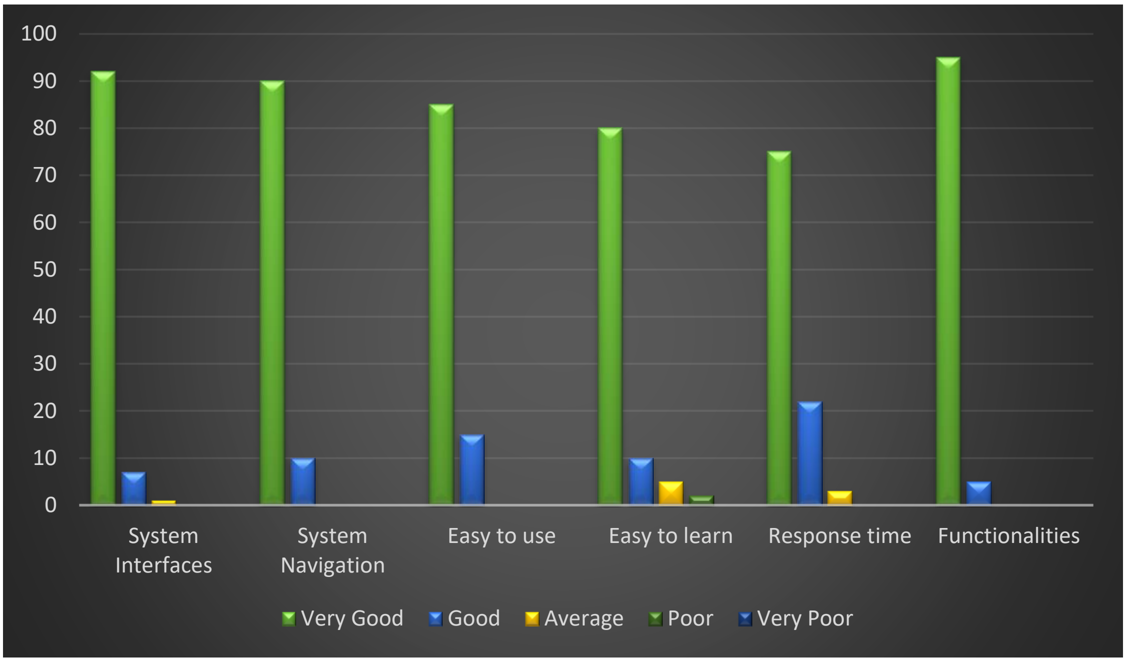


Figure 5.1: The summery of the user feedback

CHAPTER 6 – CONCLUSION

6.1 OVERVIEW

St. Joseph’s Balika Vidyalaya is a leading girls’ school which is situated in the city of Kegalle. The management authority and the teachers of the school always looking forward to provide a quality and better service for achieving the school vision and mission success. In order to do so as the main extra-curricular activity, Sports becomes a major factor.

Sports’ related activities of the school has been handled and maintain manually. The need of an information system arises in such an environment that it is hard to maintain all the details manually and it was time consuming.

To overcome these disadvantages, they help me to build this new system. Top level management and other system users get huge support to maintain sports activities effective and efficiency way.

6.2 LESSONS LEARNT

- The individual project was a great opportunity to get a practical knowledge of what we have been learnt throughout the BIT degree programe.
- The project was the great option to understand SDLC and each phase in system development life cycle.
- Time management was important task and learnt to manage time with specified time frame
- It was great opportunity to study the PHP server-side scripting language, JavaScript client-side validations and other tools and technologies which used to develop the system.
- Learnt how to communicate with system users and get information from them through friendly and formal manner.
- It was important to fix the errors when they found while developing, since the small errors could affect other developments on later stages.

6.3 FUTURE IMPROVEMENTS

Some suggestions for improvements in the future are as follows.

- Providing the SMS facility after processing special tasks.
- Develop a chat facility to communicate among system users.
- Develop the system as mobile application
- Provide a facility to do payments for the Coaches via online.

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APPENDIX A - SYSTEM MANUAL

This section of the documentation provides necessary information to the system users to install and configure the system in technical perspective.

HARDWARE REQUIREMENTS

Hardware	Recommended Minimum Requirements
Processor	Pentium(R) 4 CPU
Memory	512MB RAM or more
Hard disk	80 GB
Display	1024x768 or resolutions above high color 16-bit display
Printer	Inkjet or LaserJet Printer
Internet	ADSL Connection (Minimum speed 512 kbps)

Table A.1: Hardware Requirements

SOFTWARE REQUIREMENTS

Software	Recommended minimum requirements
Operating System	Windows XP/ Vista/ Windows 7/ Windows 8/Windows 10
XAMPP	Xampp-v3.2.4, Apache server, MySQL
Web Browser	Internet Explorer/ Google chrome/ Fire fox
Code Editor	NetBeans IDE 11.2 (Need for future developments)

Table A.2: Software Requirements

SOFTWARE INSTALLATION

Step 01: Installing XAMPP

- Download XAMPP from http://www.apachefriends.org for windows and install (refer Table A.2 for the minimum version). Install the XAMPP into C:\xampp of the computer.
- Run the XAMPP control panel. XAMPP control panel is shown in Figure A.1.

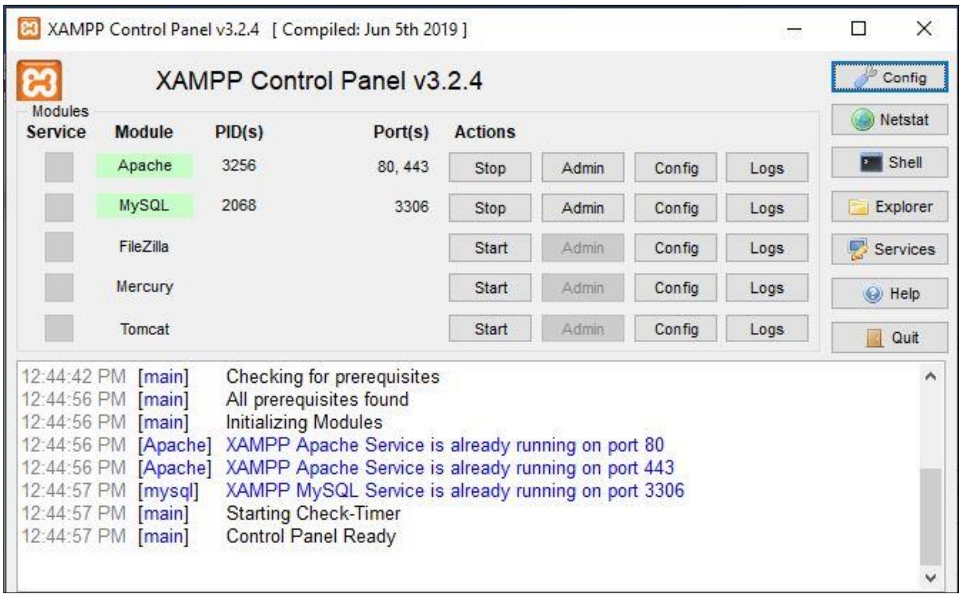

Figure A.1 XAMPP Control Panel

Step 02: Database Installation

- Click on “Admin” in mysql .
- Go to Databases tab and type a database name as “db_ssmgtss”and click on create button.
- Navigate to the “Import” tab and click “choose file” button.
- Browse the CD and select the “ssmgtsys.sql” file by opening Database folder.
  OR

- Open the web browser and type the URL http:/localhost/phpmyadmin/ in the address bar and enter username and the password (if you set username and password)

- Create new empty database named ‚db_ssmgtsys‟

- Navigate to the “Import” tab and click “Choose File” button to browse the database file.

- Select CD and open the Database folder from browse window. Select “ssmgtsys.sql” file in the Database folder

- Press “GO‟ button which located in the bottom of the page.

Step 03: Files Extraction

- Open the CD and copy the system folder “SSMgtS” and paste it to the dictionary path “C:\ xampp \ htdocs”.

Step 04 (Final Step): Run the System

- Open the web browser and type URL as “http:// localhost/ SSMgtS” and then click “Enter” to run the system.

APPENDIX B - DESIGN DOCUMENTAION

Use Case Narratives

Use case Create Game

Use case number	UC-02	
Use-case Name	Create games	
Priority	High	
Actor	Principal	
Description	This use case permits Principal to add games into the system.	
Pre-condition	UC-01	
Post condition	Games are successfully added to the system.	
Basic course of Action	User Action	System Response
	1. Principal wants to create a new game. 2. Principal requests the page of create game. 4. Principal enters the following information into the form. (Game, Game description) 5. The Principal clicks on the ‘Create Game’ Button.	3. The system displays the form to be filled about information of new game. 6. The system verifies that all the fields are filled. 7. The system verifies that the filled data is valid. 8. System displays a message as ‘Game was created Successfully.’ 9. Use case Exit
Alternate course of Action	6.1 If one or few fields are empty the system back to basic course of action 4 and display a message as ‘<Field> should not be empty.’ 7.1 If the entered information is invalid the system back to basic course of action 4 and display a message as ‘<Field> is invalid’.	

Table B.1 Use case Create Game

Use case Add Game Details.

Use case number	UC-03	
Use-case Name	Add Game Details	
Priority	High	
Actor	PTI	
Description	This use case permits PTI to add events and their information to the System.	
Pre-condition	UC-01, UC-02	
Post condition	Games and Athletics events information are successfully added to the system.	
Basic course of Action	User Action	System Response
	1. The PTI wants to add events into the system. 2.PTI requests the page of adding events. 4. PTI enters the following information into the form. (Game, Event, Event Type [Individual/ Double/ Team], no of team members, MC requirement) 5. The PTI clicks on the 'Add Event' Button.	3. The system displays the form to be filled about information of events. 6. The system verifies that all the fields are filled. 7. The system verifies that the filled data is valid. 8. System displays a message as 'Event was added Successfully.' 10. Use case Exit
Alternate course of Action	6.1 If one or few fields are empty the system back to basic course of action 4 and display a message as '<Field> should not be empty.' 7.1 If the entered information is invalid the system back to basic course of action 4 and display a message as '<Field> is invalid'.	

Table B.2 Add Game Details

Use case Apply as a Coach

Use case number	UC-05	
Use-case Name	Apply as a Coach.	
Priority	Medium	
Actor	Coach	
Description	This use case permits Coach to apply as a Coach.	
Pre-condition	None	
Post condition	Coach successfully applied for a game as a coach.	
Basic course of Action	User Action	System Response
	1. A Coach wants to apply for a game for the post of sport's Coach. 2.Actor requests the page of 'Apply as a coach'. 4. The Coach enter the following information into the form. (Date, Register year, Title, Name with initials, Address, Mobile Number, Email address, NIC Number, Photograph, Highest Edu. Qualifications, Professional Qualifications, Experiences as a Coach,) 5. The Coach clicks on the 'Apply' Button.	3.The system displays the form to be filled to apply as a Coach. 6. The system verifies that all the fields are filled. 7. The system verifies that the filled data is valid. 8. System displays a message as 'You applied Successfully.' 10. Use case Exit
Alternate course of Action	6.1 If one or few fields are empty the system back to basic course of action 4 and display a message as '<Field> should not be empty.' 7.1 If the entered information is invalid the system back to basic course of action 4 and display a message as '<Field> is invalid'.	

Table B.3 Use case Apply as a Coach

Use Case Schedule Coach Interview

Use case number	UC-06	
Use-case Name	Schedule Coaches' interviews.	
Priority	High	
Actor	Principal	
Description	This use case permits Principal to schedule Coaches' interviews.	
Pre-condition	UC- 01, UC-05	
Post condition	Principal successfully schedules Coaches' interviews.	
Basic course of Action	User Action	System Response
	1. The Principal wants to schedule Coaches' interviews. 2.Actor requests the page of 'Schedule Interviews'. 3. Actor clicks on the 'View Applicants' button. 5. The Principal click on an applicant. 7. The Principal enters the following information to the form. (Date, Time, Req. Docs) 8.The PTI clicks on 'Ok' button.	4.The system displays applicants for games. 6. The system displays a form to enter schedule information. 9. The system verifies that all the fields are filled. 10. The system verifies that the filled data is valid. 11. System completes schedule and displays a message as 'Interview schedule is successfully completed.' 12. The system send the schedule to relevant applicant via email. 15. Use case Exit
Alternate course of Action	10.1If one or few fields are empty the system back to basic course of action 4 and display a message as '<Field> should not be empty.' 10.1 If the entered information is invalid the system back to basic course of action 4 and display a message as '<Field> is invalid'.	

Table B.4 Use case Schedule Coach Interview

Use case Add Upcoming Tournament Details

Use case number	UC-13	
Use-case Name	Add upcoming tournament details.	
Priority	High	
Actor	PTI	
Description	This use case permits PTI to add upcoming tournament details to other users' reference.	
Pre-condition	UC-01	
Post condition	Successfully add upcoming tournament details.	
Basic course of Action	User Action	System Response
	1. PTI wants to add upcoming tournaments details. 2. PTI requests the page. 4. PTI enters the following information to the form. (Tournament name, Relevant Game, Start date, End Date, Upload Calling Letter) 5. Coach clicks on 'Add' button.	3.The system displays the form to add upcoming tournament details. 6. The system verifies that all the fields are filled. 7. The system verifies that the filled data is valid. 8. Use case Exit
Alternate course of Action	6.1If one or few fields are empty the system back to basic course of action 4 and display a message as '<Field> should not be empty.' 7.1If the entered information is invalid the system back to basic course of action 4 and display a message as '<Field> is invalid'.	

Table B.4 Add Upcoming Tournaments

APPENDIX C

USER DOCUMENTATION

User Documentation is helped to users to access the system and use it easily. This User Documentation contains with main components of the system.

User Login

Figure C.1 is shown the System login interface for all users. Only authorized users can access into the system by entering valid username and password. If invalid user name or password is entered system will display an error message.

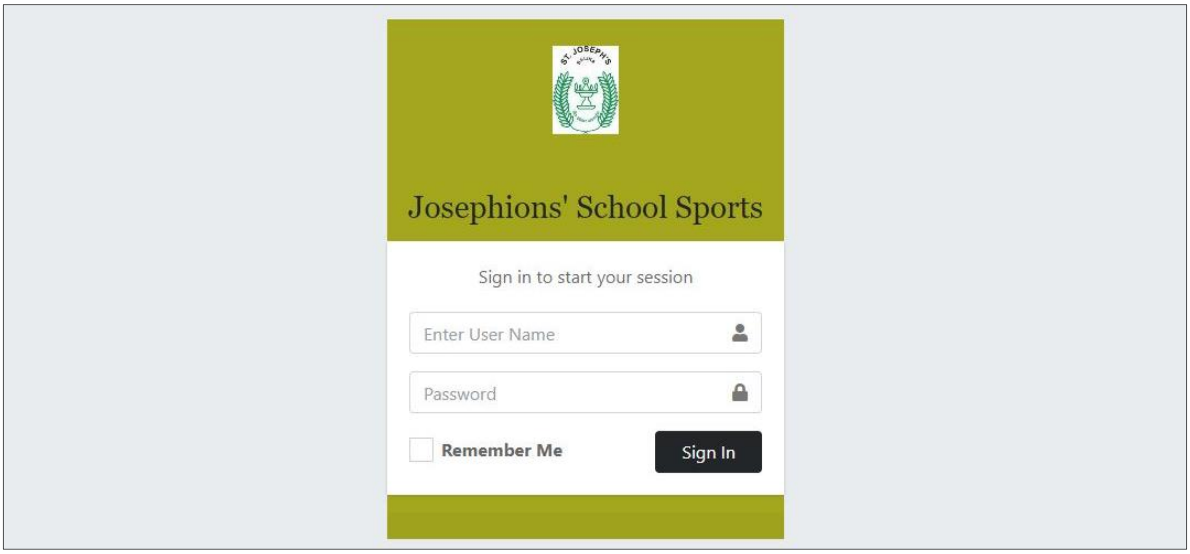

Figure C.1 User Login

User Dashboard

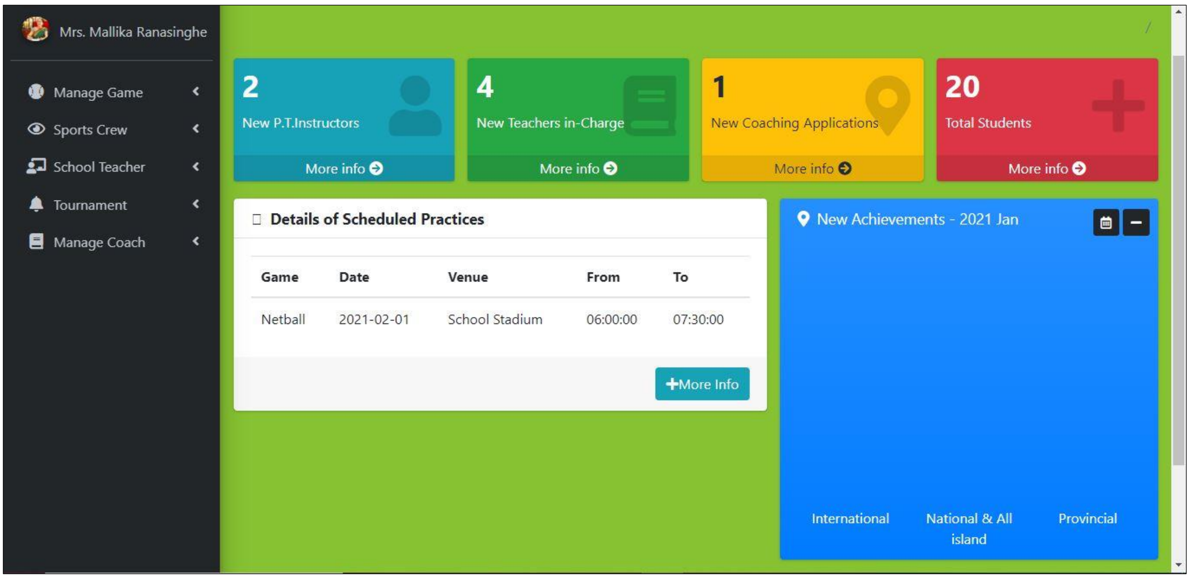

Figure C.2 Principal Dashboard

Add Games

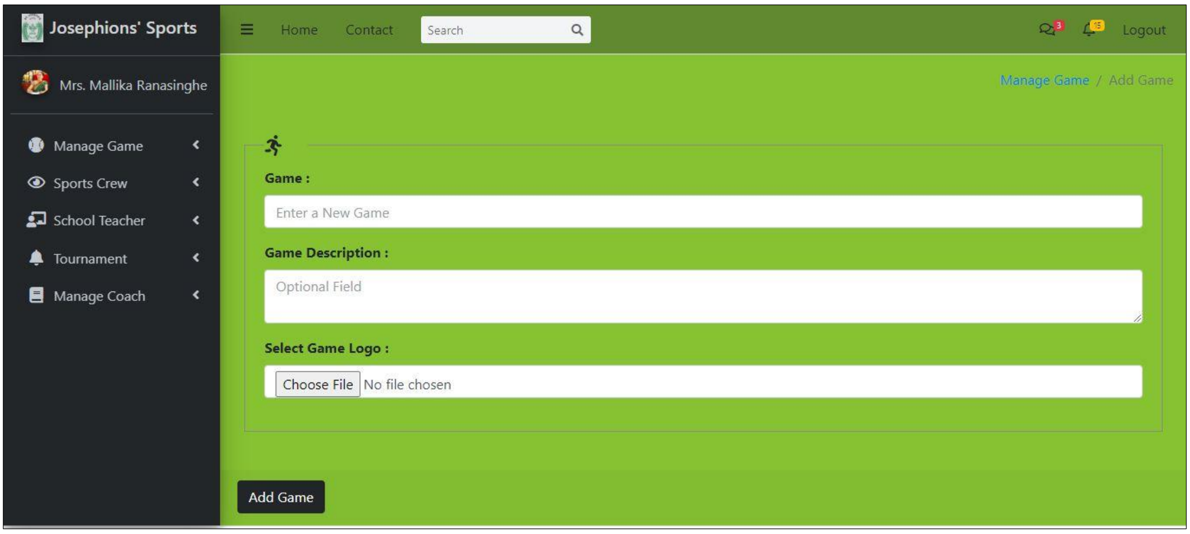

Figure C.3 Add Games

View Users

Figure C.4 View Applied Coach

Assign Games to Sport Staff

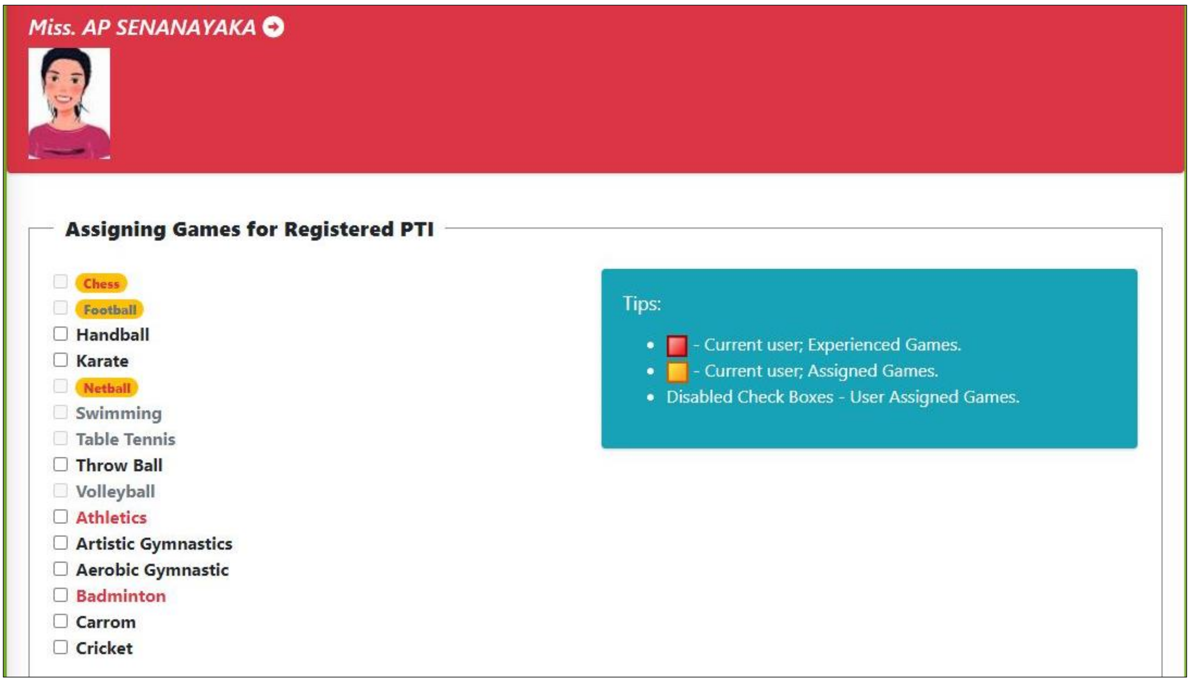

Figure C.5 Assign Games for PTI

View Details

Figure C.6 View Tournament Details

Schedule Activities

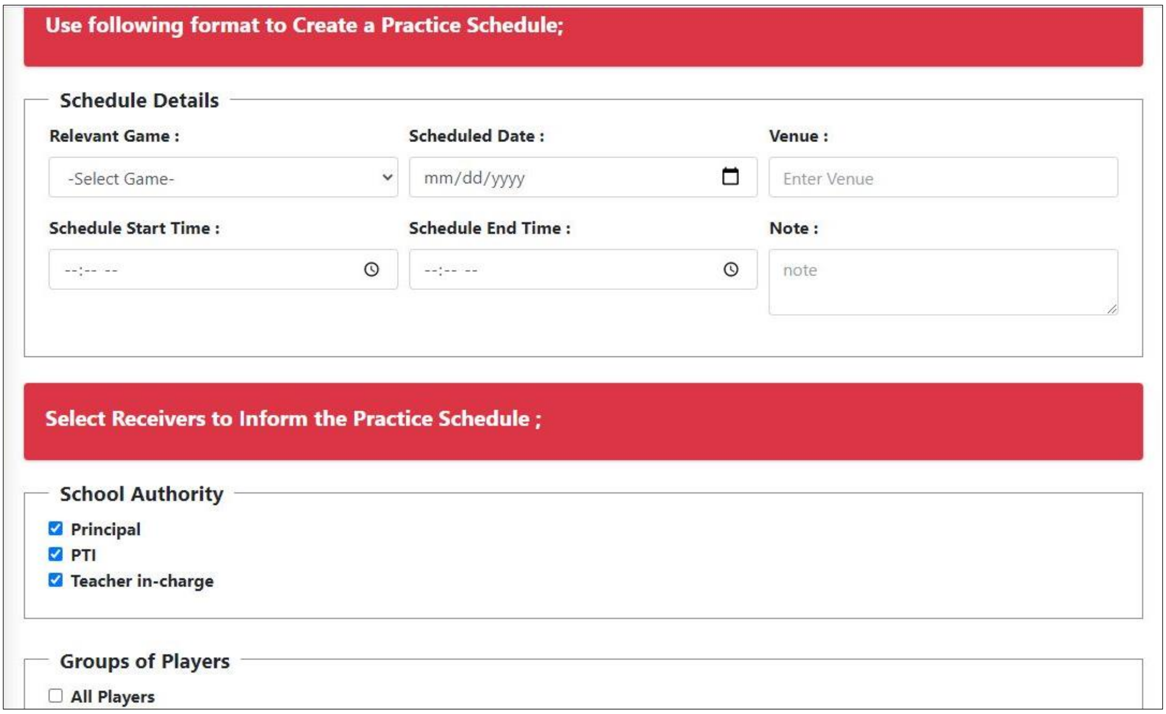

Figure C.7 Practice Scheduling

Attendance Marking

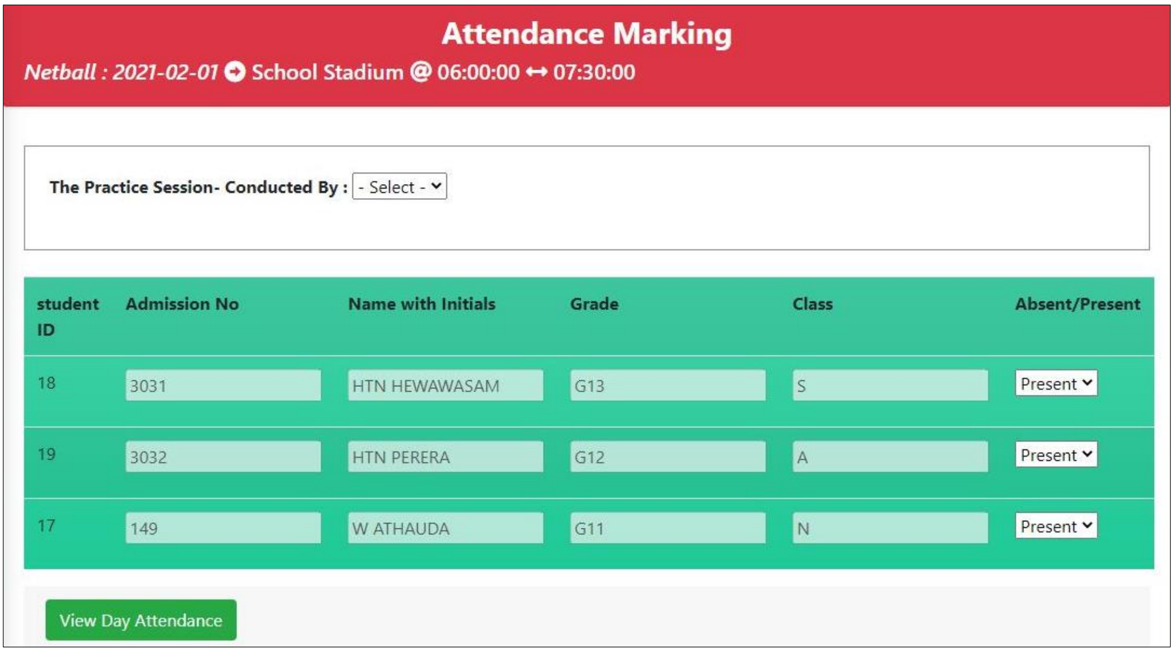

Figure C.8 Attendance Marking

Body Mass Index

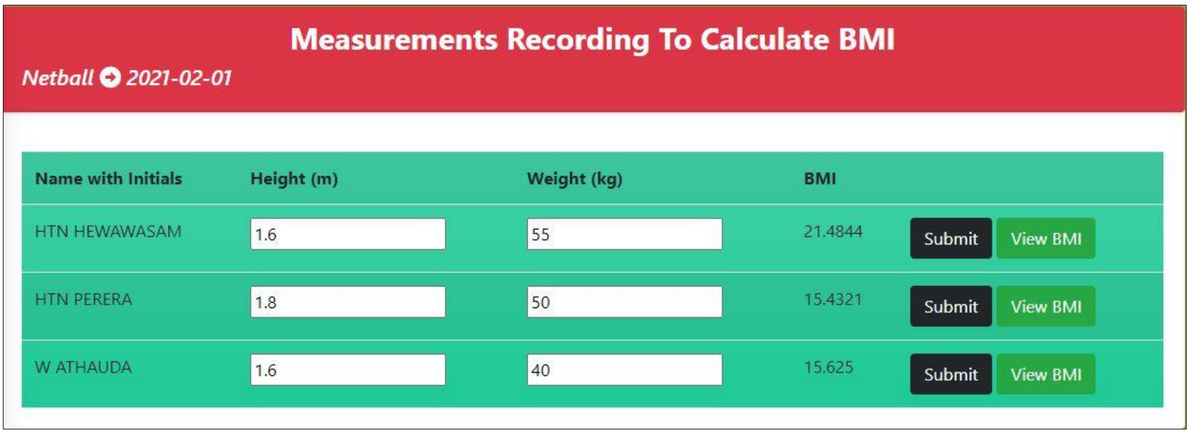

Figure C.9 Calculate BMI

Skill Test for Athletes

Figure C.10 Skill Test Navigation Screen

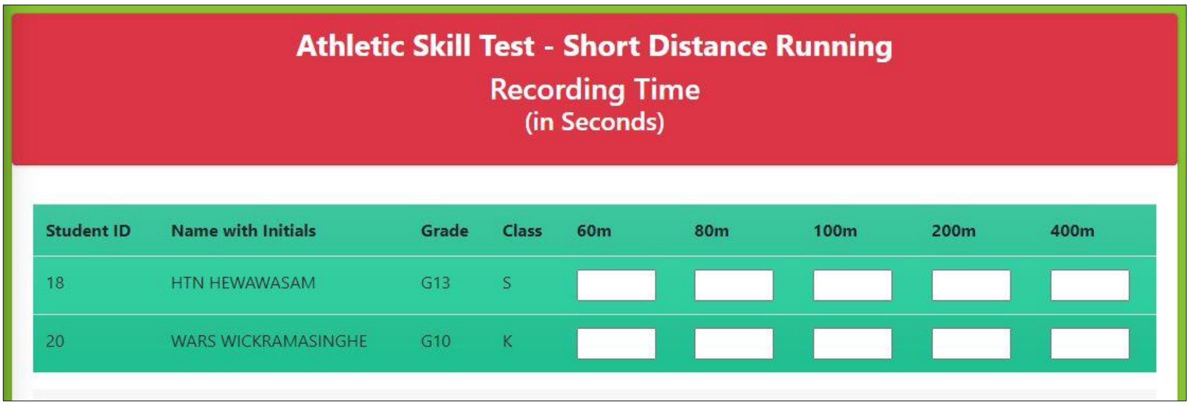

Figure C.11 Skill Test – Short Distance Running

APPENDIX D-MANAGEMENT REPORTS

APPENDIX E-TEST RESULTS

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